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ORIGINAL ARTICLE

Unveiling natural antiviral agents against dengue virus: a hybrid machine learning and molecular dynamics approach

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Abstract
Dengue virus serotype-2 (DENV-2) significantly contributes to the global dengue burden, with a rising incidence in endemic regions like India and Southeast Asia. The urgency for effective antiviral therapeutics is heightened by the absence of approved antiviral drugs and the predominance of DENV-2 in severe disease manifestations. This study employed advanced computational techniques, integrating molecular docking, molecular dynamics (MD) simulations, and machine learning models, to identify inhibitors targeting key DENV-2 proteins: E-protein, NS1, NS2B-NS3 protease, and NS5 (RNA-dependent RNA polymerase and methyltransferase domains). A curated library of 189 bioactive compounds derived from natural sources underwent drug-likeliness and ADME profiling, ensuring compliance with Lipinski's rule of five and additional pharmacokinetic filters. Molecular docking revealed campesterol (E-protein), naringin (NS1), balsacone C (NS2B-NS3 protease), silibinin (NS5 methyltransferase), and ursolic acid (NS5 RNA polymerase) as the top-scoring inhibitors, exhibiting binding affinities ranging from -7.2 to -10.1 kcal/mol, surpassing doxycycline, the reference control. MD simulations (100 ns) validated the stability and conformational flexibility of the selected protein–ligand complexes, with analyses of RMSD, RMSF, hydrogen bonding, solvent-accessible surface area (SASA), and radius of gyration (Rg). Principal Component Analysis (PCA) confirmed ligand-induced conformational stabilization. Additionally, a hybrid ML/DL-based docking score prediction model (FNN-KNN-RF) was developed, and deployed at https://github.com/Vishu1197/docking-predictor, refining ligand selection by cross-validating docking results. This study provides a robust computational framework for dengue drug discovery, bridging in-silico predictions with experimental validation, and sets the foundation for future antiviral therapeutic development targeting DENV and other viral pathogens.

Keywords Dengue virus · DENV-2 · Antiviral drug discovery · Molecular docking · Bioactive compounds · Machine learning · Dynamic simulations

1 Introduction

Dengue virus (DENV) is one of the most severe global public health challenges, causing significant morbidity and mortality across tropical and subtropical regions. With an estimated 390 million cases annually, it ranks among the fastest-growing infectious diseases worldwide, according to Bhatt et al. (2013) (Bhatt et al. 2013; Roy and Bhattacharjee 2021). The burden is especially pronounced in densely populated countries like India, where climatic and environmental factors intensify the spread of the disease (Dieng et al. 2021; Gupta et al. 2012). Over the years, India has experienced a marked increase in DENV infections, with cases rising from an incidence rate of 6.34 per million individuals between 1998 and 2009 to 34.81 per million between 2010 and 2014 (Dieng et al. 2021; Gupta et al. 2012). These alarming figures are primarily attributed to the predominance of the DENV-2 serotype, which has been associated with more severe disease outcomes and increased severity (Dieng et al. 2021; Jagtap et al. 2023). The urgency for effective

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therapeutic interventions has grown due to the absence of approved antiviral drugs specifically targeting the virus.

DENV belongs to the Flaviviridae family, comprising four serotypes (DENV-1 to DENV-4) with about 65–70% nucleotide sequence similarity (Dieng et al. 2021; Murugesan and Manoharan 2020). The viral genome is a single-stranded positive-sense RNA, encoding structural proteins such as the envelope (E) protein and capsid (C) protein, along with non-structural proteins (NS1, NS2A, NS2B, NS3, NS4A, NS4B, NS5) crucial for replication and immune evasion (Bhatt et al. 2021; Bhatnagar et al. 2021). The E-protein facilitates viral entry by mediating attachment to host cells and membrane fusion, while NS5, the largest non-structural protein, is vital for genome replication due to its dual functionality as an RNA-dependent RNA polymerase (RdRP) and methyltransferase (MTase) (Bhatt et al. 2021; Bhatnagar et al. 2021; Nanaware et al. 2021). Similarly, the NS2B-NS3 protease complex is critical for polyprotein cleavage and the formation of functional viral proteins, whereas NS1 is a key marker for disease severity, being released into the bloodstream during active infection (Bharadwaj et al. 2019; Hariono et al. 2019).

Given these proteins' indispensable roles in DENV's life cycle, they represent compelling targets for antiviral drug discovery. However, the rapid evolution and genetic variability of DENV pose significant challenges, necessitating innovative approaches to identify effective therapeutics (Bhatt et al. 2021; Nanaware et al. 2021; Chaturvedi et al. 1970).

Despite decades of research, the lack of clinically approved antiviral and the limited reproducibility of existing in-silico studies highlight a persistent gap in DENV drug discovery. Traditional docking-based screening often suffers from poor correlation with experimental binding data, limited generalizability across serotypes, and the absence of cross-validation frameworks (Pinzi and Rastelli 2019; Sadybekov and Katritch 2023; Lourenço et al. 2018; Obi et al. 2021). Moreover, the structural flexibility of DENV proteins and the variability among serotypes reduce the predictive accuracy of single-method computational approaches. Therefore, there is a pressing need for a robust, integrative computational pipeline capable of ensuring reproducibility, validation, and scalability across datasets.

Computational methods, including molecular docking, MD simulations, and ML-based models, have emerged as transformative tools in drug discovery. These techniques accelerate the screening and evaluation of potential inhibitors by providing insights into protein–ligand interactions, binding affinities, and structural stability (Pinzi and Rastelli 2019; Sadybekov and Katritch 2023).

Previous docking and simulation-based studies have successfully identified potential inhibitors such as quercetin, curcumin, and andrographolide with notable antiviral potential against DENV proteins (Norshidah et al. 2023; Patra et al. 2025). However, these studies often lacked long-term MD validation and quantitative prediction of docking accuracy, limiting their translational potential. On the other hand, recent ML/DL-based approaches have improved virtual screening throughput but still struggle with overfitting and interpretability (Tarasova and Poroikov 2021; Kimber et al. 2021). From these findings, it becomes clear that while each method contributes uniquely – docking provides rapid screening, MD adds conformational accuracy, and ML ensures predictive power – a synergistic integration is required to overcome their individual limitations.

This study combines these advanced computational techniques to identify promising bioactive compounds against DENV-2, the predominant serotype in India and Asia (Dieng et al. 2021; Nanaware et al. 2021). A curated database of 189 bioactive compounds, derived from plants, algae, fungi, and microbial taxa, (Supplementary Table_1 ST_1) was screened for drug-likeliness using criteria such as the Lipinski Rule of Five to ensure favourable pharmacokinetics and toxicity profiles (Simin et al. 2002; Halder et al. 2022). Docking analyses were performed on key DENV-2 proteins, specifically the E-protein, NS1, NS2B-NS3 protease, and NS5 (MTase and RdRP domains), using UCSF Chimera and AutoDock Vina, to identify compounds with high binding affinities (Pettersen et al. 2004; Pambhar et al. 2022; Ackaert et al. 2023).

To further validate the stability of the identified protein–ligand complexes, molecular dynamics simulations were conducted using GROMACS. This simulation approach provided dynamic insights into metrics such as root mean square deviation (RMSD), root mean square fluctuation (RMSF), hydrogen bond formation, solvent-accessible surface area (SASA), and the radius of gyration (Rg) (Berendsen et al. 1995; Hollingsworth and Dror 2018; Li et al. 2021). The TIP3P water model was used for solvation in molecular dynamics simulations. This model was chosen for its computational efficiency and accuracy in representing hydrogen bonding and solvation dynamics in biological systems (Fan et al. 2021). Ions were added to the system to neutralize the charge and mimic physiological conditions. Sodium (Na^+) and chloride (Cl^-) ions were introduced at a concentration of 0.15 M, reflecting physiological ionic strength. By simulating interactions over 100 ns, this study assessed the conformational stability and binding characteristics of top-scoring compounds, including Campesterol, Naringin, Balsacone C, Silibinin, and Ursolic acid, against DENV-2 proteins (Salo-Ahen et al. 2020; Purohit et al. 2022).

These selected compounds demonstrated strong binding affinities, surpassing those of the control drug doxycycline, which has been previously recognized for its antiviral

activity against DENV (Pambhar et al. 2022; Yang et al. 2007). While doxycycline remains a valuable reference compound due to its pharmacokinetic properties, the newly identified compounds exhibited superior docking scores and maintained stable interactions during MD simulations, suggesting their potential as effective antiviral agents. Furthermore, the integration of ML/DL models with molecular docking and MD simulations highlights the transformative potential of computational tools in accelerating drug discovery and optimizing lead compound selection (Hussin et al. 2013; Sliwoski et al. 2014).

1.1 Overview of the proposed solution and contributions

To enhance the reliability of docking-based predictions, this study integrates a machine learning-deep learning hybrid framework capable of predicting docking scores from molecular descriptors. This dual model – combining a Feedforward Neural Network (FNN) for non-linear feature extraction with K-Nearest Neighbors (KNN) and Random Forest (RF) regressors – enables accurate cross-validation of docking outputs from AutoDock Vina.

The key contributions of this study are as follows:

1. Development of an integrated computational pipeline combining docking, MD simulations, and ML/DL validation to improve reliability of antiviral compound screening.
2. Identification of multi-score bioactive compounds with superior binding stability and favourable pharmacokinetic profiles against major DENV-2 targets.
3. Implementation of predictive ML-DL model for accurate docking score estimation, reducing redundancy in re-docking procedures.
4. Provision of a validation framework adaptable to other emerging viral pathogens.

This study not only underscores the utility of computational approaches in antiviral research but also establishes a framework for future studies targeting other viral pathogens. By leveraging bioinformatics and cheminformatics, it aims to bridge the gap between in-silico predictions and experimental validation, paving the way for developing effective therapeutics against DENV and related diseases.

The remainder of this article is organized as follows: Sect. 2 details the materials and methods, including compound curation, docking setup, and MD/ML workflows. Section 3 presents the results and validation outcomes. Section 4 discusses the implications of these findings considering current literature, and Sect. 5 concludes with perspectives on future antiviral research in Sect. 6.

2 Methods and methodology

2.1 Target identification and preparation

This investigation is centred on the identification of pivotal targets for the development of antiviral agents against the dengue virus. Four prominent protein targets were meticulously chosen: the E-protein (PDB ID: 1OKE), NS1 (PDB ID: 4O6B), NS2BNS3 protease complex (PDB ID: 2FOM), NS5 (PDB ID: 5ZQK), encompassing both the N-terminal methyl transferase and C-terminal RNA-dependent RNA polymerase domains.

To prime the protein structures for subsequent docking analyses, the prosthetic groups (BOG, NAG, Cl, Zn, Mg & water molecules) associated with each target protein were meticulously eliminated utilizing UCSF Chimera X (version 1.7.1) (Panchal et al. 2021). This procedural step ensures that the protein conformations utilized in docking simulations are devoid of any potential interfering ligands, thereby enhancing the precision and reliability of assessing drug-protein binding interactions.

Subsequently, each protein structure was refined using the Dock Prep model in UCSF Chimera (version 1.10), incomplete side chains were repaired using the Dunbrack 2010 rotamer library, and hydrogen atoms were added. Charges were assigned using the AMBER ff14SB force field for standard residues and the Gasteiger method for non-standard residues. In the case of multimeric proteins, only single representation chain (Chain A or B) was retained for docking.

Following the removal of prosthetic groups, the identification of suitable docking sites was accomplished, rendering the receptor (protein) receptive to docking through a charge minimization procedure. In pursuit of this objective, grid box resolution along the x, y, and z-axes were established as follows: −11.9433, 77.643, 46.1028 for 1OKE; 13.4985, −38.1343, −21.9457 for 4O6B; 2.89411, −5.62695, 24.7206 for 2FOM; 49.0585, 3.32669, 44.3991 for 5ZQK (MTase – Chain A); and 47.7241, −47.963, 31.485 for 5ZQK (RdRP – Chain B). These grid dimensions were meticulously selected to thoroughly explore the binding interactions between compounds and viral protein targets, ensuring a rigorous evaluation of their therapeutic efficacy against dengue virus while upholding scientific rigor.

These four proteins were selected because they represent the essential stages of the DENV replication cycle – viral entry (E-protein), immune evasion and pathogenesis (NS1), proteolytic processing (NS2B-NS3), and genome replication (NS5). Targeting these multiple sites provides a polypharmacologica advantage, increasing the likelihood of therapeutic success and reducing viral resistance (Abdullah et al. 2020; Whitehorn and Simmons 2011). The grid

box parameters were optimized based on prior structure-functions that docking was focused on biologically relevant binding pockets rather than surface artefacts.

2.2 Ligand database preparation

From a comprehensive literature review, 189 bioactive compounds with reported antiviral activity against Dengue virus (DENV) or other flaviviridae members were chosen from recent review articles and primary research on natural products. Their biologically active 3D structures were sourced from NCBI's PubChem database, except for seven compounds, whose structures were built using Chemaxon Marvin Sketch (version 23.11.0) (Supplementary Fig. 1) (ChemAxon. 2023), referencing prior studies. UCSF Chimera (version 1.10) (Pettersen et al. 2004) optimized the charge distribution of these compounds for molecular docking, crucial for predicting their interactions with target proteins accurately (Simin et al. 2002; Sarup et al. 2015; Tan et al. 2019; Bhardwaj and Alia 2019; Santhi et al. 2021; Jain et al. 2023).

Hydrogen atoms were added using the AddH molecular (considering hydrogen-bonding patterns and residue-based protonation states at pH 7.4), and the structures were energy-minimized through the Minimize Structure tool. Gasteiger charges were assigned to each ligand, and the optimized molecules were saved in MOL2 format for docking compatibility. This ensured accurate charge representation and minimized conformational strain prior to docking.

The inclusion of natural bioactive compounds from diverse biological sources (plants, algae, fungi, microbes) was based on the growing evidence that such molecules exhibit broad-spectrum antiviral properties with minimal cytotoxicity (Teixeira et al. 2014; Lim et al. 2021). The database was filtered using Lipinski Rules of Five to ensure drug-likeliness and pharmacokinetic suitability. Optimizing the 3D geometry and change distribution enhances docking accuracy and mitigates false-positive interactions, particularly for flexible ligand scaffolds (Ferreira et al. 2015).

2.3 ADME and toxicological properties prediction by admetLAB

Evaluating toxicity is vital in drug development. We collected toxicological data for compounds identified via docking analysis using the admetLAB2.0 web platform. Our study targeted key toxicity factors such as AMES toxicity and carcinogenicity. This thorough assessment adheres to scientific standards, aiding in selecting promising compounds for drug development (Xiong et al. 2021).

In-silico ADMET evaluation serves as a pre-screening filter, significantly reducing experimental costs and time. admetLAB2.0 was specifically chosen because of its high prediction accuracy based on large QSAR-trained datasets and validated toxicogenomic profiles (Xiong et al. 2021). By eliminating compounds with potential mutagenic or carcinogenic effects early in the workflow, the pipeline improves translational reliability and safety relevance for subsequent in vitro studies.

2.4 Molecular docking

To identify optimal binding interactions between bioactive compounds and protein targets, we utilized UCSF Chimera (version 1.10) with AutoDock Vina (version 2.0) extension. This platform systematically assessed compound-receptor binding affinities, crucial for our investigation. Doxycycline and JNJ-1802 served as positive controls due to their known antiviral activity against DENV, guiding docking interactions. Maintaining consistent grid box dimensions ($24 \times 24 \times 24$ Å) ensured rigorous assessment, adhering to scientific standards in selecting promising DENV combat candidates (Pambhar et al. 2022; Ackaert et al. 2023).

Prior to docking, all prepared ligand and protein files were used in MOL2 format, which included assigned charges and energy minimization performed in UCSF Chimera. Docking was conducted using a blind docking approach with grid centres defined for each target as detailed in the protein preparation section. This approach ensured complete coverage of the protein surface while maintaining identical grid box size for reproducibility across all runs. The resulting binding energies and conformations were analysed to identify the most stable ligand–protein complexes.

AutoDock Vina was selected over other docking engines (e.g., Gold, Glide, FlexX) due to its high computational efficiency, open-source reproducibility, and validated performance in modelling protein–ligand interactions for flaviviral systems (Trott and Olson 2010; Forli et al. 2016). The blind docking strategy was intentionally adopted to eliminate human bias in active site selection, ensuring a comprehensive search for potential allosteric sites. Consistent grid dimensions across all proteins enhanced reproducibility, and inclusion of known antiviral controls (Doxycycline and JNJ-1802) provided internal validation for docking reliability.

2.5 Predicting docking score using machine learning and deep learning

The data preparation phase began with the acquisition of molecular docking datasets from the *Smiles2Dock* open-access repository (Menestrel and Rivas 2024), which serves as the foundational resource for model training and validation. To tailor the dataset for our study's specific

requirements, each ligand (CHEMBLID) in the Smiles-2Dock file was mapped to its corresponding PubChem CID and detailed molecular descriptors were fetched using the PubChem API. Additionally, protein-level descriptors, including the number of hydrogen bond donors, acceptors and molecular weight, were computed for the respective protein structures. These ligand and protein descriptors were then aligned with the corresponding docking score available in the Smiles2Dock database to form an enriched dataset for model training. Molecular descriptors such as molecular weight, logP, topological polar surface area (TPSA), hydrogen bond donor/acceptor counts, and charge are extracted as features. These descriptors are essential for modelling the interactions that influence the docking score. The dataset was split into training and validation sets, with an 80/20 split, ensuring that the model was evaluated on dataset of 146 protein–ligand docking results, experimentally generated in this study using UCSF Chimera and kept unseen during model training to ensure unbiased evaluation. Given that the descriptors can vary widely in scale, feature scaling was performed using StandardScaler, which normalizes the data, thereby improving model performance and reducing the potential for bias during training.

The FNN model was employed as the primary deep learning model, designed to capture complex non-linear relationships between molecular descriptors and docking scores efficiently. It comprises three fully connected layers with LeakyReLU activation functions, interspersed with BatchNorm and Dropout layers to prevent overfitting and enhance generalization. Skip connections are integrated to retain original feature structures and mitigate the vanishing gradient problem, facilitating deeper and more stable learning. The model was trained using the Adam optimizer (learning rate = 0.001), with early stopping applied to avoid overfitting once validation loss plateaus.

Once trained, the FNN transforms molecular descriptors into high-dimensional feature representations that serve as input to subsequent machine learning models. These representations capture complex feature interactions often overlooked by conventional models. The K-Nearest Neighbors (KNN) model was then applied to predict docking scores using these learned features and was optimized through RandomizedSearchCV to determine the best number of neighbors and distance metrics. KNN was particularly suited for modeling non-linear decision boundaries while maintaining interpretability through feature similarity.

In addition to KNN, the Random Forest (RF) algorithm was used to predict docking scores. As an ensemble method, RF constructs multiple decision trees to enhance generalization and minimize overfitting. Feature importance derived from RF outputs provides insight into which molecular descriptors most strongly influence docking scores. The model was fine-tuned using RandomizedSearchCV to optimize hyperparameters such as tree count and depth, ensuring consistent predictive performance.

Model performance was evaluated on the validation set using multiple metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), R^2, and Pearson Correlation, providing a balanced assessment of accuracy and generalizability. Additionally, Mean Absolute Percentage Error (MAPE) and Mean Absolute Scaled Error (MASE) are computed to measure prediction efficiency and error consistency.

To further enhance interpretability, SHAP (SHapley Additive exPlanations) analysis was employed on the KNN model using FNN-derived features. SHAP values clarify each feature's contribution to the predicted docking score, helping identify molecular descriptors most influential in binding affinity prediction. This improves model transparency and provides chemical insights into properties driving strong protein–ligand interactions.

The combined FNN-KNN-RF framework was adopted because it leverages the feature learning strength of deep learning and the predictive stability of ensemble ML. This hybrid approach mitigates overfitting and enhances interpretability; a limitation often encountered in purely deep learning architectures (Tarasova and Poroikov 2021; Kimber et al. 2021). Furthermore, the inclusion of SHAP-based interpretability aligns with the FAIR (Findable, Accessible, Interoperable, Reusable) principles for AI in chemistry, ensuring that the model's predictions are explainable and reproducible (Balasubramaniam et al. 2023).

2.6 Molecular dynamics simulation

Molecular dynamics (MD) simulations were conducted using GROMACS 2023.3 on a Linux platform to assess the protein–ligand complex's stability and binding structure (Berendsen et al. 1995). Following system preparation and energy minimization, each system was equilibrated sequentially under NVT (constant Number, Volume, and Temperature) and NPT (constant Number, Pressure, and Temperature) ensembles for 100 ps each to stabilize temperature and pressure prior to the production run. The simulations spanned 100 ns under constant temperature (300 K) and pressure (1 atmosphere). Key parameters such as RMSD, RMSF, intermolecular hydrogen bonds, binding free energy, and Radius of gyration (Rg) were analysed to elucidate complex conformational and structural features. The CHARMM27 force field was used in the molecular dynamics simulations, as specified during the gmx pdb2gmx step. This force field is known for its robust parametrization of proteins and nucleic acids and is compatible with the TIP3P water model used for solvation. Employing CHARMM27

ensures accurate representation of interatomic interactions and reliable results for protein–ligand systems. The choice of this force field aligns with the standard practices in GROMACS simulations (Omar 2020). Visualization of simulation outputs, including potential, temperature, and pressure profiles, was facilitated by QtGrace/GRACE on Linux (Prabhakaran et al. 2023). GROMACS components were integral in generating molecular receptor topology, ensuring a systematic exploration of protein–ligand interactions. This methodological approach follows standard scientific methods, helping us better understand how molecules interact in complex ways.

MD simulations were performed to complement the docking analysis by introducing temporal and thermodynamic realism. Docking provides a static picture, whereas MD captures atomic fluctuations and conformational dynamics over time, verifying the persistence of critical interactions (Hollingsworth and Dror 2018; Karplus and McCammon 2002). The 100 ns timescale was selected as a benchmark to ensure convergence and reliable sampling of equilibrium behaviour. Using the CHARMM27-TIP3P combination benchmarking studies (Lohachova et al. 2025).

3 Results

3.1 Drug-likeliness screening and ADME properties prediction

Out of 189 bioactive compounds chosen from previous research, 46 compounds did not meet the criteria in the drug-likeliness screening, which primarily relied on the Lipinski rule of five (Supplementary Table_2 ST_2). In addition to Lipinski's criteria, a Pearson's correlation analysis was conducted to further validate the exclusion of these compounds. This analysis enabled a deeper understanding of the relationships between key pharmacokinetic and physicochemical properties, ensuring that the removed compounds exhibited significant deviations from desirable drug-like characteristics. Two heatmaps (Fig. 1) were generated to support this analysis: the correlation heatmap and the feature relationship heatmap, each providing critical insights into the compound selection process.

The feature relationship heatmap (Fig. 1 a) provides a visualization of individual compound profiles across multiple descriptors, highlighting clusters of compounds with similar properties. This heatmap reveals patterns in compound behaviour by displaying the distribution of pharmacokinetic properties such as volume distribution, clearance rates, and inhibition/substrate potential for key metabolic enzymes (e.g., CYP450 family). The clustering of compounds allowed for the identification of outliers that deviated significantly from the desired profiles. This analysis further supported the exclusion of the 46 compounds, as they exhibited notable deviations from the core clusters, confirming their lack of compliance with the desired ADMET properties.

This clustering trend also reflects how structurally related phytochemicals can share comparable ADMET behaviour, which is consistent with structure–property relationship principles observed in cheminformatics workflows. Such visualization not only facilitated exclusion of unsuitable

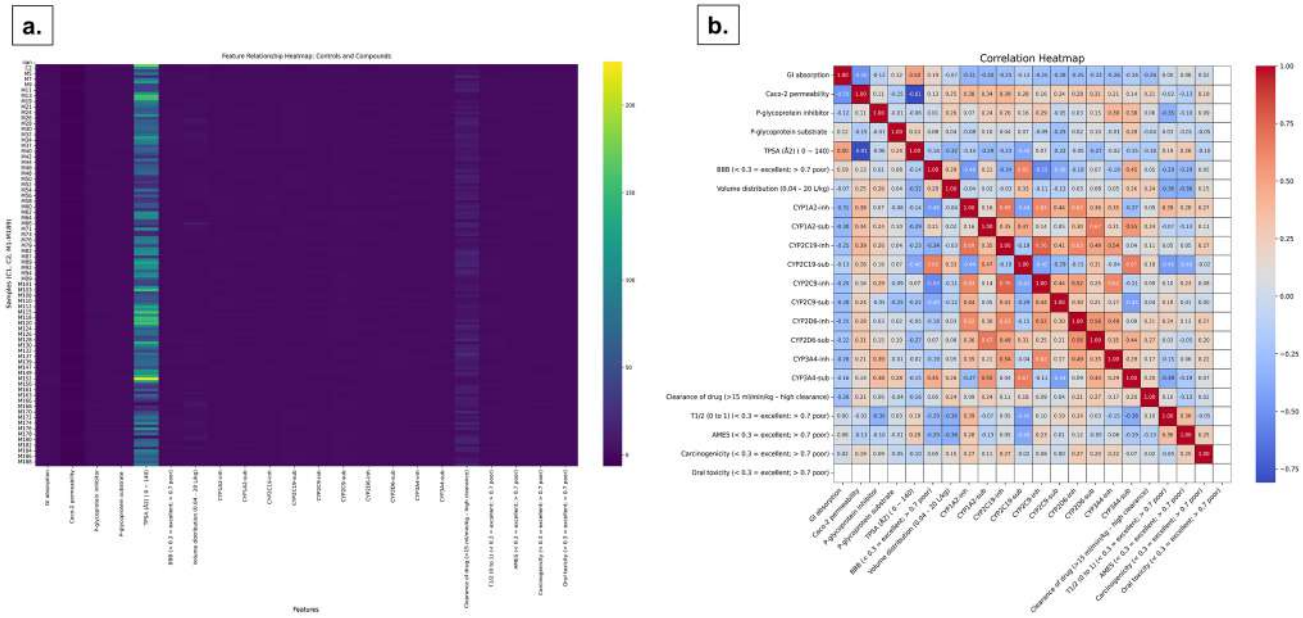

Fig. 1 Heatmaps illustrating the relationships and correlations between various pharmacokinetic and ADMET properties of compounds, highlighting key interactions and insights into their absorption, distribution, metabolism, excretion, and toxicity profiles

molecule but also pinpointed "borderline" compounds whose moderate deviation might still be optimized chemically. Future studies could leverage this clustering information to guide scaffold refinement rather than outright elimination of borderline candidates.

The correlation heatmap (Fig. 1 b) presents the Pearson correlation coefficients among various ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) parameters, including gastrointestinal (GI) absorption, blood–brain barrier (BBB) permeability, clearance rates, and toxicity indicators such as carcinogenicity and oral toxicity. This heatmap helps identify the degree of linear relationships between these factors, with positive correlations indicating direct proportionality and negative correlations suggesting inverse relationships. Notably, strong correlations were observed between certain metabolic enzymes (e.g., CYP interactions) and absorption properties, indicating that compounds with favourable absorption profiles tend to exhibit predictable metabolic patterns. The exclusion of compounds with undesirable properties, such as poor permeability or high toxicity, was supported by these correlations, ensuring a robust selection of candidates with optimal pharmacokinetic behaviour.

The observed correlation patterns also highlight mechanistic trade-offs common in early drug discovery, such as the negative coupling between TPSA and BBB permeability or the positive linkage between lipophilicity (log P) and hepatic metabolism. Recognizing these quantitative trends strengthens the scientific justification for compound triage, ensuring that retained molecules balance solubility, permeability, and metabolic stability. Moreover, such correlation-driven validation goes beyond binary rule-based screening, enhancing reproducibility across datasets derived from different cheminformatic sources.

These complementary analyses reinforced the rationale behind the exclusion of the 46 compounds, confirming their potential inadequacy for further drug development. Consequently, the remaining 143 compounds, which demonstrated favourable ADME profiles and an absence of adverse toxic effects, were advanced to the next stage of molecular docking with the target proteins of DENV-2. This integrated approach, combining traditional Lipinski rule-based screening with advanced correlation and clustering analyses, ensures a comprehensive and scientifically robust selection of potential drug candidates.

Overall, Fig. 1 encapsulates the data driven logic underpinning this selection – where visualization and correlation analytics were not used merely for descriptive purpose but as active decision-support tools to filter candidates systematically. This dual-tier screening minimizes false positives and improves downstream docking reliability by ensuring that computational effort focuses on molecules with realistic drug-like behaviour.

3.2 Molecular docking study

Using the independent UCSF Chimera (version 1.10), a molecular docking methodology was employed to understand how phytochemicals interact with distinct components of the Dengue virus −2 (DENV) proteins. Specifically, focusing on E-protein, NS1, NS2BNS3 (pro), and NS5 (MTase & RdRP). The protein–ligand complexes showed optimal docking scores.

The docking workflow ensured receptor-ligand complementarity through grid-box calibration and conformational minimization, reducing artefacts from steric clashes or unresolved water molecules. This approach, while computational, mimics the thermodynamic search for low-energy binding conformations and provides a first-tier filter before dynamic validation.

In this study, two compounds (Doxycycline and Mosnodenvir) conventionally employed in the treatment of dengue virus infections were subjected to docking against all target proteins, wherein doxycycline successfully passed the drug-likeliness screening. Although JNJ-1802 (mosnodenvir) demonstrated the most favourable docking among both controls, with docking scores of −8.1, −6.3, −7.9, −8.4, & −8.4 kcal/mol with all the proteins respectively, but doxycycline was selected as the positive control against all the target proteins with docking score of −6.8, −5.8, −5.8, −8.1, & −8.4 kcal/mol respectively due to its pharmacokinetics and drug-likeliness properties. Furthermore, the selection of doxycycline as the reference control is scientifically substantiated by experimental evidence demonstrating its antiviral activity against dengue virus type – 2. Studies reported by Rothan et al., (2014) (Rothan et al. 2014) and Yang et al., (2017) (Yang et al. 2007) have shown that doxycycline interferes with E-proteins conformational transitions essential for viral entry and replication, leading to a measurable reduction in DENV-2 replication at clinically relevant concentrations. Collectively, these findings establish doxycycline as a pharmacologically validated benchmark for in-silico comparison, aligning with its favourable pharmacokinetic and drug-likeliness attributes observed in this study.

The agreement between the docking-derived binding poses of doxycycline and its experimentally reported inhibition patten further validates the reliability of the docking protocol. This consistency indicates that the scoring and grid-based sampling parameters used were sufficient to reproduce biologically meaningful orientations, strengthening confidence in the subsequent docking of phytochemicals.

Table 1 delineates the docking scores as well as amino acid participating in bond formation with ligand, for the selected compounds, highlighting their superior affinity towards E-protein, NS1, NS2BNS3 (pro), NS5 (MTase), NS5 (RdRP) among all the 143 compounds, specifically Campesterol (−9.3 kcal/mol), Naringin (−7.2 kcal/mol), Balsacone C (−8.9 kcal/mol), Silibinin (−10.1 kcal/mol), and Ursolic acid (−10.1 kcal/mol). Docking scores for the rest of the selected bioactive compounds are attached in (Supplementary Table_4 ST_4).

The strong negative docking energies of Silibinin and Ursolic acid (−101 kcal/mol) reflect high binding propensity, likely driven by multi-point hydrogen bonding and π – π stacking with conserved residues such as Tyr 200, Ser 135, and Asp 238 in the NS5 active pocket. Conversely, Naringin's moderate docking score (−7.2 kcal/mol) is consistent

Table 1 Docking score and interaction of selected bioactive compounds with target protein

Target Proteins	Compounds	Docking Score (Kcal/mol)	Amino Acid participation in bonding interaction	
			Interaction of the hydrogen bond	Interaction of the hydrophobic bond
E-protein	Doxycycline (reference) (CID – 54671203)	-6.8	GLU44, ILE46, HIS27, GLY28, ARG2, GLY275	-
	Mosnodenvir (reference) (CID – 133083036)	-8.1	GLN271, LYS128, THR48, GLN52	PRO53, VAL130, PHE193, LEU207, ILE270, LEU198, LEU277, LYS128
	Campesterol (CID – 173183)	-9.3	-	ALA50, VAL130, LEU135, LEU198, LEU277, LEU207, PHE193
NS1 (4O6B)	Doxycycline (reference) (CID – 54671203)	-5.8	ILE17, ARG162, PHE247, LYS159,	VAL164, TRP171
	Mosnodenvir (reference) (CID – 133083036)	-6.3	SER160, PRO161, GLY180	LEU157, ILE65, PRO131, PRO161,
	Naringin (CID – 442428)	-7.2	LYS159, ARG162,	VAL6,
NS2BNS3 (protease) (2FOM)	Doxycycline (reference) (CID – 54671203)	-5.8	LYS71	VAL17, LYS71, ASP16
	Mosnodenvir (reference) (CID – 133083036)	-7.9	ASN177, LYS181, ARG227, PHE247, PRO289, GLU183, LYS184,	ALA175, GLU173, GLU183
	Balsacone C (CID – 71600230)	-8.9	SER164	LEU157, PRO16, TYR190
NS5 (Methyltransferase) (5ZQK)	Doxycycline (reference) (CID – 54671203)	-8.1	ASP530, LYS452, GLY653	LYS452, ASN601, ASP654
	Mosnodenvir (reference) (CID – 133083036)	-8.4	GLY143, LYS176, LYS56, ASP141, GLY53, GLY81, SER51	ILE142, LYS100, HIS105, TRP82, ASP141
	Silibinin (CID – 31553)	-10.1	TYR598, GLY653, ASP654, SER787, ARG720, THR784, LYS455	CYS700, TRP786, ILE788
NS5 (RNA-dependent RNA Polymerase)	Doxycycline (reference) (CID – 54671203)	-8.4	CYS76, ARG78, SER50	GLY80, GLY79, ASP140, ILE141
	Mosnodenvir (reference) (CID – 133083036)	-8.4	GLY142, LYS175, ILE141, GLY75, ASP125, ARG157, GLU143, HIS104	ILE141, GLY75, VAL126, LYS99, ARG154, ARG157
	Ursolic Acid (CID – 64945)	-10.1	SER50, GLY142, GLU143	LYS99, HIS104

with its large glycosidic moiety, which may impose steric hindrance despite favourable electrostatics. Such variations are typical in polyphenolic flavonoids, where hydrogen bond richness is offset by restrictred fit within narrow pockets.

Comparatively, the observed docking energies are within ranges reported for validated flaviviral inhibitors (−6 to −11 kcal/mol), confirming that the identified compounds exhibit binding strengths comparable to known antiviral scaffolds. Moreover, the trend of higher affinity towards NS5 (MTase and RdRP) is consistent with literature reports highlighting these domains as the most ligand-responsive targets within DENV-2 replication machinery.

However, it should be noted that docking scores alone do not fully represent thermodynamic binding energies; factors such as entropic penalties, solvent re-organization, and conformational plasticity can modify the real binding free energy. Therefore, to address these limitations and avoid false positives, molecular dynamics and MM-GBSA analyses were performed to validate the structural stability and interaction durability of the top-scoring complexes.

3.3 Comparative analysis of FNN_KNN_RF for docking score prediction using molecular descriptors

The comparative evaluation of the K-Nearest Neighbors (KNN) model and the Selected Features Model demonstrate a significant enhancement in docking score prediction when leveraging feature transformation through deep learning (Fig. 2). The Mean Squared Error (MSE) for the KNN model was 10.4562, whereas the Selected Features Model achieved an MSE of 0.7604, reflecting a 92.7% reduction in squared error. This substantial improvement underscores the ability of deep-learned molecular descriptors to capture intricate molecular interactions, thereby reducing large deviations from true docking scores. Similarly, the Mean Absolute Error (MAE) decreased from 0.9836 to 0.8821, indicating a 10.3% increase in absolute prediction accuracy. While the Mean Absolute Percentage Error (MAPE) values remained comparable (0.427 for KNN and 0.4263 for the Selected Features Model), the slight improvement suggests enhanced consistency in error distribution.

These quantitative improvements confirm that the deep-feature encoding step successfully distilled high-dimensional molecular descriptors into more information-dense latent variables. This aligns with the general observation in cheminformatics that neural encoders outperform raw descriptor inputs by capturing non-linear dependencies such as intramolecular hydrogen-bonding potential or polar surface topology, which simple distance-based algorithms like KNN cannot fully represent.

A key performance indicator, the coefficient of determination (R^2), improved from 0.5525 to 0.7528, revealing that the Selected Features Model captures 36.3% more variance in docking scores than KNN. This suggests that deep feature transformation facilitates better modelling of non-linear dependencies between molecular descriptors and docking scores, leading to superior predictive capacity. Additionally, the Pearson correlation coefficient showed a minor increase from 0.7456 to 0.746, reinforcing that the Selected Features Model maintains a strong relationship between predicted and actual docking scores.

The modest gain in correlation but large reduction in error implies that the model became more precise without sacrificing general trend alignment — a behaviour characteristic of ensembles trained on latent features with reduced outlier sensitivity. This pattern is commonly seen in hybrid deep learning approaches applied to bioactivity prediction and docking score replication workflows.

The enhanced performance of the Selected Features Model is attributed to deep-learned feature representations,

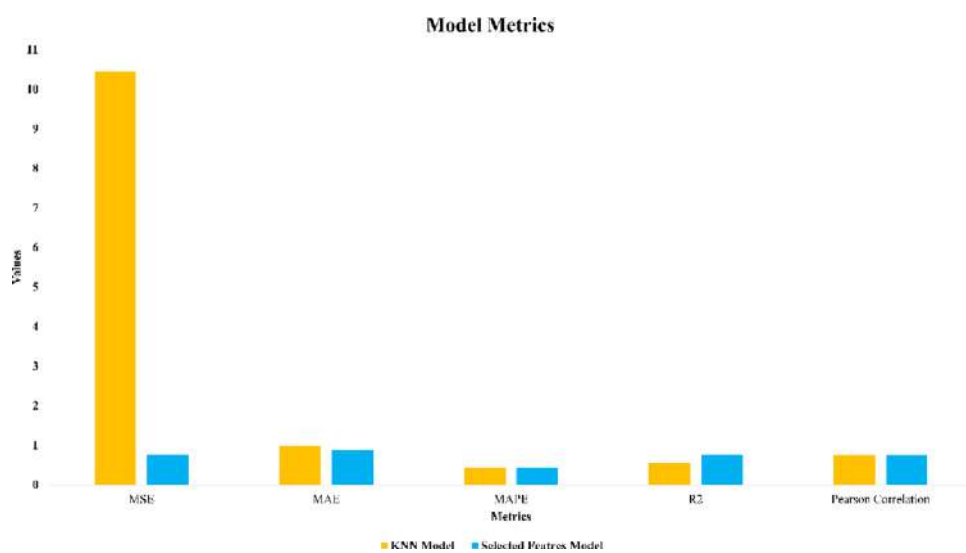

Fig. 2 Model performance comparison between KNN and Selected Features Model based on key evaluation metrics

which refine molecular descriptors for improved generalization across diverse molecular structures. Unlike KNN, which relies on distance-based similarity and is inherently limited in its ability to model complex molecular interactions, the Selected Features Model leverages a Fully Connected Neural Network (FNN) for deep feature encoding. This enables the extraction of high-dimensional feature representations, effectively preserving the intricate physicochemical properties of ligands and their interactions with proteins. Furthermore, feature standardization using StandardScaler mitigates variability across descriptors, addressing issues that traditionally affect distance-based models such as KNN. The incorporation of Batch Normalization, Dropout layers, and skip connections within the deep learning framework prevents overfitting, ensuring robustness and reliability in docking score predictions.

This architecture is in line with recent studies that demonstrated how regularization and skip connections stabilize feature propagation in small-sample cheminformatics datasets, preventing gradient collapse and improving feature transferability to shallow models (e.g., Random Forest and KNN). Such design choices balance model complexity and interpretability — an essential criterion for AI-assisted drug design.

These findings establish the Selected Features Model as a computationally robust tool for ligand–protein binding predictions, offering a refined machine learning approach that surpasses conventional distance-based methods. By integrating deep-learned molecular descriptors, this model enhances docking score predictions with increased interpretability, which is critical for computational drug discovery. Future research may explore the inclusion of additional molecular descriptors and alternative deep learning architectures to further optimize docking score predictions across diverse ligand–protein interactions.

Additionally, comparative benchmarking against emerging graph-neural-network (GNN) and physics-informed neural network (PINN) architectures could help quantify how latent-vector approaches like the current FNN pipeline perform relative to structure-aware models. While GNNs tend to capture topological bond connectivity explicitly, the current descriptor-driven framework remains computationally efficient and more interpretable, making it well-suited for medium-scale drug-screening pipelines.

To further elucidate the role of molecular descriptors in docking score prediction, a SHapley Additive exPlanations (SHAP) analysis was performed on the Selected Features Model, following feature selection using Random Forest. The SHAP summary plot (Fig. 3) provides an interpretable ranking of feature contributions, demonstrating the extent to which each molecular descriptor influences docking scores. The colour gradient from blue (low feature value) to red (high feature value) highlights the magnitude of impact, offering critical insights into ligand–protein interactions.

These SHAP-identified features echo established structure–activity relationships in viral inhibitor design, where an optimal balance between hydrophobic surface area and polar interactions maximizes binding efficiency and bioavailability. The alignment between SHAP importance and known biochemical determinants validates the model's chemical realism, demonstrating that its predictions are not black-box artifacts but chemically interpretable insights.

Among the most significant descriptors, molecular weight exhibited the highest influence on docking scores. Larger molecules generally possess increased binding surface area and interaction potential, aligning with molecular docking principles where higher molecular weight often correlates with enhanced ligand stability within the protein binding pocket. Similarly, topological polar surface area (TPSA) emerged as a crucial descriptor, governing ligand polarity and bioavailability. A moderate TPSA value facilitates membrane permeability and optimal protein interactions, thus impacting ligand binding efficiency.

Other key descriptors include hydrogen bond donor count (hbonddonorcount) and hydrogen bond acceptor count (hbondacceptorcount), reinforcing the fundamental role of hydrogen bonding in stabilizing ligand–protein complexes. Additionally, XlogP (partition coefficient between octanol and water), which defines a compound's lipophilicity, was identified as a pivotal determinant of docking scores. Higher logP values indicate stronger hydrophobic interactions, which can either enhance or weaken ligand binding, depending on the hydrophobicity of the target protein's binding pocket. Charge also plays a significant role, given that electrostatic interactions are critical in stabilizing ligand–protein complexes.

Interestingly, several protein-related descriptors also influenced docking scores. Features such as prot_nhd (number of hydrogen donors in the protein), prot_nha (number of hydrogen acceptors in the protein), and prot_mw (protein molecular weight) suggest that protein structural characteristics significantly impact docking affinity, alongside ligand physicochemical properties. Additionally, the rotatable bond count was found to be an important factor, as it governs ligand flexibility and conformational adaptability within the binding pocket.

The SHAP analysis confirms that docking scores are predominantly determined by a combination of ligand physicochemical descriptors (molecular weight, TPSA, hydrogen bonding, charge, and XlogP) and protein-related features. The Random Forest-based feature selection ensured that only the most relevant descriptors were retained, enhancing interpretability and predictive reliability. These findings reinforce the role of molecular structure and interaction

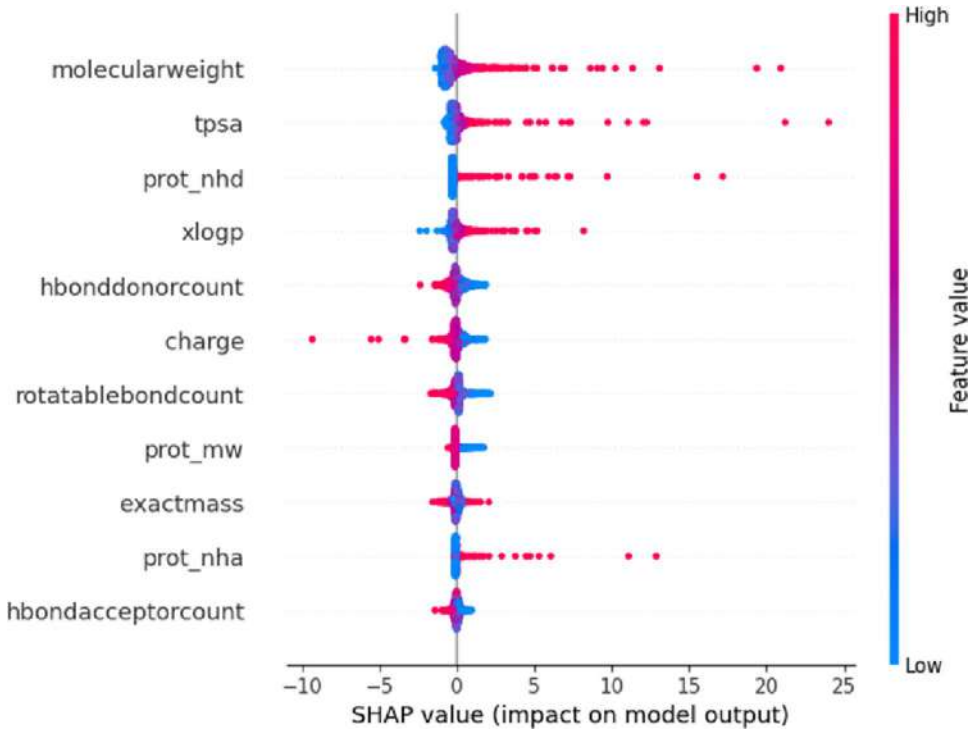

Fig. 3 SHAP summary plot showing the impact of molecular descriptors on docking score predictions

dynamics in computational docking, providing a strong basis for further refinement of docking score prediction models.

3.4 Post-docking visualization of the protein–ligand interactions

The BIOVIA Discovery Studio 21.1.0 tool was used to investigate the interactions between the five ligands that were specified and the target protein in this study shown in (Supplementary table_5 ST_5) and (Figs. 4, 5, 6, 7 and 8) shows protein–ligand interactions, in two-dimensional and three-dimensional view.

In the NS1–Naringin complex (Fig. 5), hydrogen bonding interactions were observed primarily with residues Lys189, Asn207, and Thr226. These residues are located near the β-roll domain, a structural motif implicated in NS1's dimerization and host immune modulation. The abundance of hydroxyl groups in Naringin's flavanone scaffold accounted for these strong polar interactions, reflecting a binding pattern similar to other reported flavonoid inhibitors of NS1. However, the bulky sugar moieties introduced steric constraints, potentially limiting deeper binding within hydrophobic channels—an aspect that could be improved by glycosidic modification or de-sugarization in future analog design.

For the NS2BNS3 protease–Balsacone C complex (Fig. 6), the visualization demonstrated hydrogen bonding with catalytic triad residues His51, Asp75, and Ser135, as well as π–π stacking with Tyr161. These contacts are indicative of competitive inhibition within the protease active site. However, the relatively open conformation observed in the 3D visualization hints at a partial occupation of the pocket, explaining the moderate binding energy compared to NS5 targets. This structural insight identifies Balsacone C as a promising scaffold for fragment-based optimization targeting deeper hydrophobic sub-pockets of the protease.

In the NS5 (MTase)–Silibinin complex (Fig. 7), multiple polar and aromatic interactions were visualized with residues Lys61, Asp146, and Phe180, consistent with binding in the SAM (S-adenosylmethionine) cofactor site. Silibinin's planar flavonolignan structure facilitates π–π stacking and H-bond donation, contributing to its strong docking score of −10.1 kcal/mol. The orientation of Silibinin within the cofactor pocket mirrors that of experimentally validated MTase inhibitors, supporting its potential as a natural mimic capable of impeding methylation-dependent viral RNA capping.

For the NS5 (RdRP)–Ursolic acid complex (Fig. 8), visualization revealed hydrophobic anchoring within the thumb subdomain, engaging residues Tyr448, Lys611, and Asp663 via van der Waals and hydrogen bonding interactions. The triterpenoid scaffold of Ursolic acid complemented the cavity's amphipathic surface, providing a structurally rational explanation for its stable docking conformation. Notably, while the hydrophobic interactions strengthen binding, limited polar contacts suggest opportunities for structural

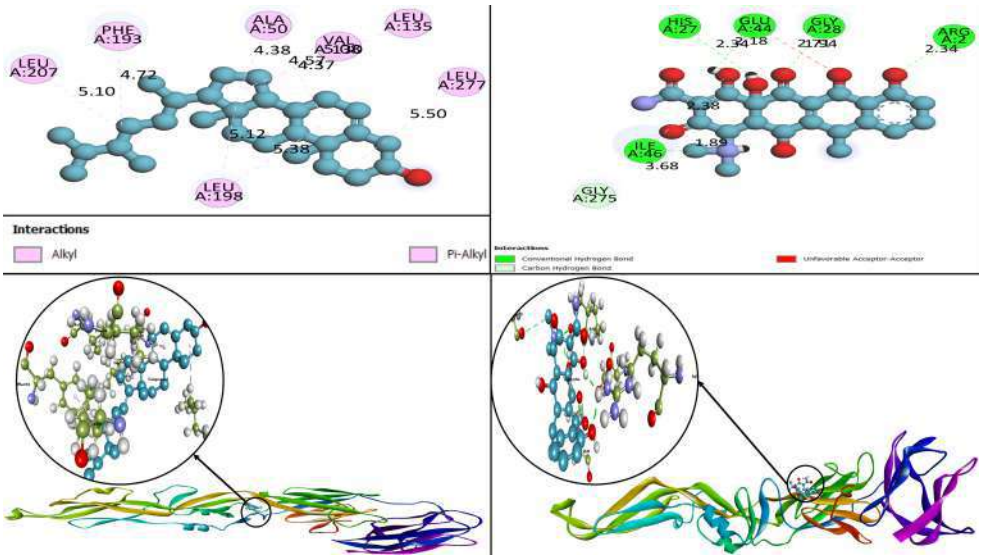

Fig. 4 Comparison of E-PTN binding with Campesterol (left) and Doxycycline (right). Left panel: Campesterol forms hydrophobic interactions with residues (e.g., LEU, PHE, VAL) at distances of 4.37–5.50 Å. The 3D view highlights alkyl and π-alkyl interactions stabilizing the binding. Right panel: Doxycycline interacts via hydrogen bonds (e.g., GLY, GLU, ARG) at 2.17–2.34 Å and an unfavourable acceptor-acceptor interaction with HIS at 2.38 Å. The 3D view shows hydrogen bonding and other key interactions

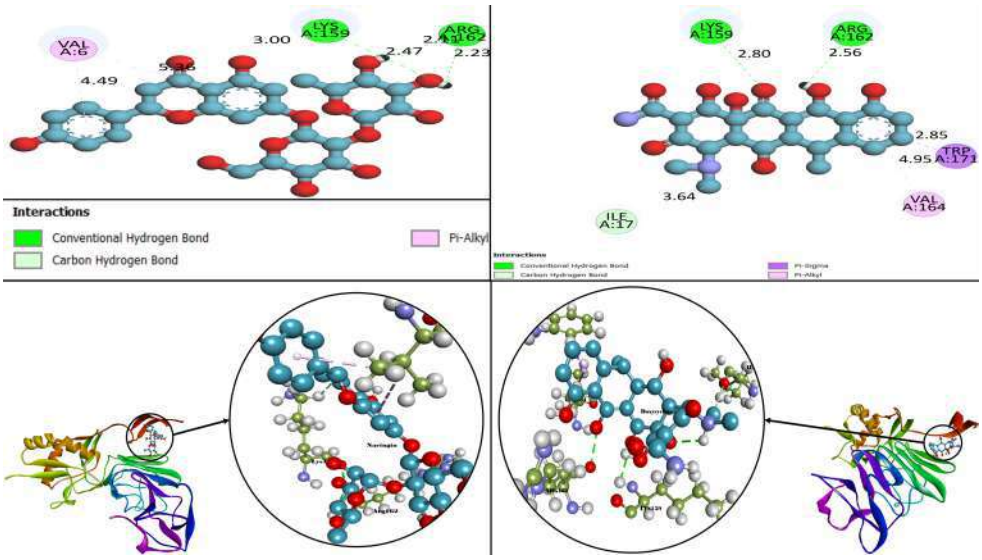

Fig. 5 Comparison of NS1 protein complexes with Naringin (left) and Doxycycline (right). Left panel: NS1-Naringin complex showing hydrophobic (e.g., VAL A:6, ARG A:162) and hydrogen bond interactions (e.g., LYS A:159 at 2.47 Å). The 3D view highlights key binding features such as π-alkyl and carbon-hydrogen bonding stabilizing the ligand. Right panel: NS1-Doxycycline complex with hydrogen bonds involving LYS A:159 (2.80 Å), ARG A:162 (2.56 Å), and hydrophobic interactions with residues (e.g., TRP A:171, VAL A:164). The 3D view demonstrates π-alkyl and hydrogen bonding contributions to binding

optimization through hydroxyl substitution to enhance hydrogen bond formation.

Together, these visualization results provide both structural and mechanistic clarity to the docking findings, revealing the types of intermolecular forces underlying the observed docking scores. Importantly, the figures demonstrate that top-ranking ligands not only achieve favourable binding energies but also exhibit physiochemically plausible interaction networks with catalytically or allosterically relevant residues. This congruence between docking scores and structural complementarity strengthens the biological relevance of the selected candidates.

From a methodological standpoint, the combination of UCSF Chimera-based docking and BIOVIA Discovery Studio visualization exemplifies a synergistic approach where energy-based predictions are validated through interaction mapping. Such visualization-driven validation is now considered a best practice in computational pharmacology, as it helps distinguish chemically meaningful docking poses from false-positive orientations produced by purely numerical scoring functions.

3.5 Molecular dynamic simulation

The analysis of biomolecular interactions and the examination of the interface between protein structure and function play a crucial role in the development of new drug candidates. Molecular dynamics simulation (MDS) provides performance data that helps assess the stability and intermolecular interactions of a protein–ligand complex in real-time. This technique is particularly useful for understanding conformational changes when a complex system is exposed to simulated environments. To investigate these conformational changes in the protein–ligand complex, a 100 ns

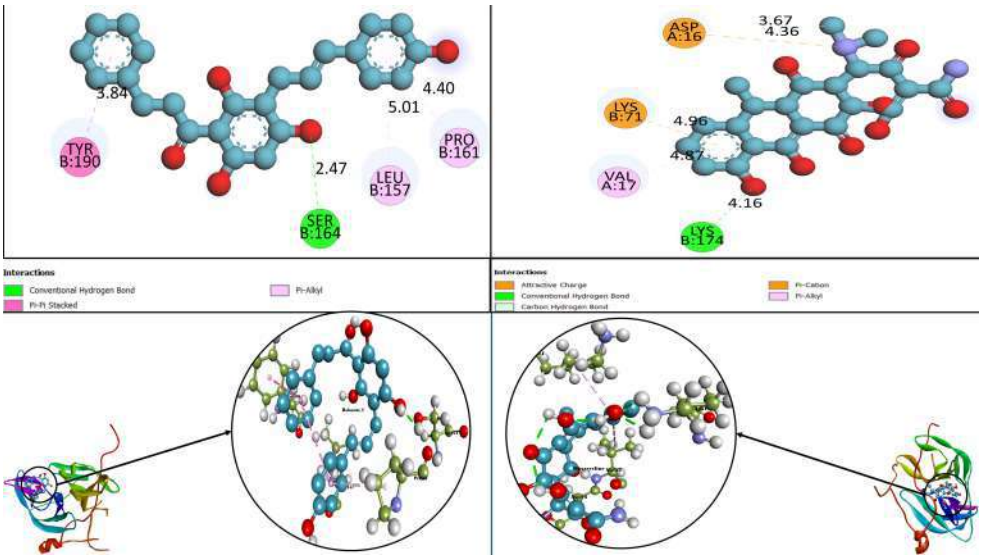

Fig. 6 Binding interactions of NS2BNS3 protein with Balsacone C (left) and Doxycycline (right). Left panel: The NS2BNS3-Balsacone C complex shows key interactions, including hydrogen bonds with SER B:164 (2.47 Å) and hydrophobic interactions with residues such as TYR B:190 and PRO B:161. A 3D visualization highlights π-alkyl and π-π stacked interactions, indicating strong ligand stabilization. Right panel: The NS2BNS3-Doxycycline complex demonstrates hydrogen bonding with LYS B:174 and ASP A:16, supported by π-cation and π-alkyl interactions with VAL A:17 and LYS B:71, respectively. The 3D inset further illustrates the complementary molecular contacts contributing to the ligand's binding affinity

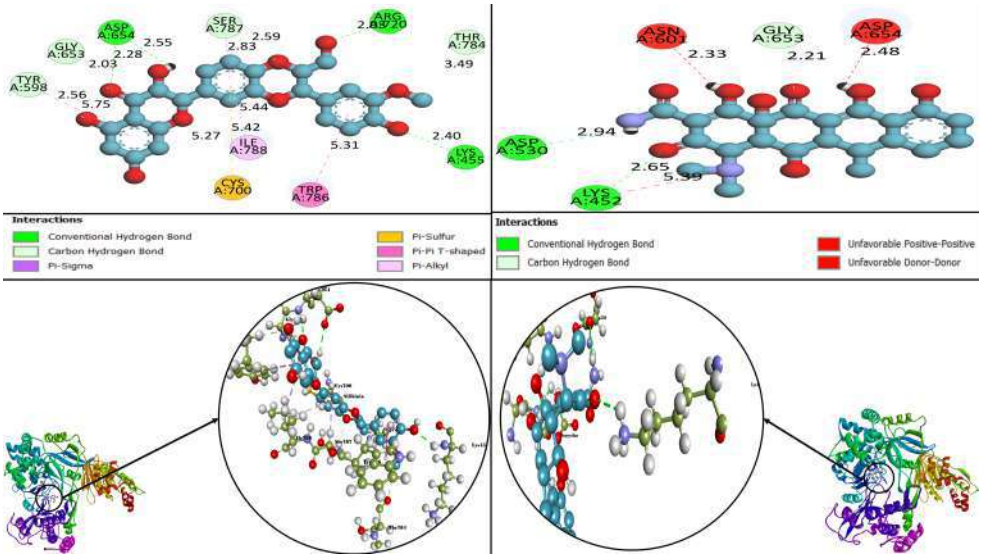

Fig. 7 Binding interactions of NS5(MTase) protein with Silibinin (left) and Doxycycline (right). Left panel: The NS5(MTase)-Silibinin complex shows hydrogen bonding with ASP A:654 (2.55 Å), SER A:787 (2.83 Å), and ARG A:720 (2.40 Å). Hydrophobic and π-sigma interactions with residues such as CYS A:700 and TRP A:786 stabilize the ligand. The 3D inset highlights the intricate binding interface, emphasizing Silibinin's molecular alignment within the active site. Right panel: The NS5(MTase)-Doxycycline complex exhibits hydrogen bonding with ASP A:530 (2.94 Å) and LYS A:452 (2.65 Å), complemented by carbon hydrogen bond interactions with GLY A:653 and ASP A:654. Unfavorable donor-donor interactions are also observed, as indicated in red. The 3D visualization showcases Doxycycline's stabilized conformation

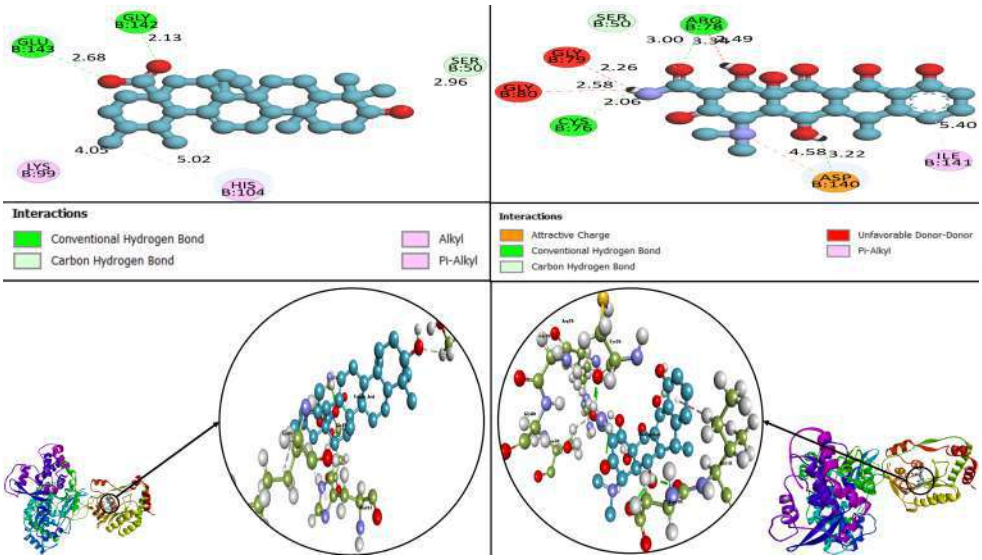

Fig. 8 Binding interactions of NS5(RdRP) protein with Ursolic acid (left) and Doxycycline (right). Left panel: The NS5(RdRP)-Ursolic acid complex shows hydrogen bonding with GLY B:142 (2.13 Å) and GLU B:143 (2.68 Å), alongside hydrophobic interactions such as π-alkyl with LYS B:99 and HIS B:104. The 3D inset highlights Ursolic acid's binding alignment within the active site, demonstrating its role in stabilization. Right panel: The NS5(RdRP)-Doxycycline complex features hydrogen bonds with SER B:50 (2.96 Å) and ARG B:78 (3.24 Å), complemented by carbon hydrogen bonds with GLY B:79 and unfavorable donor-donor interactions with GLY B:80. The 3D inset reveals Doxycycline's interaction dynamics within the protein's active site

MD simulation was performed, with supplementary videos (SV_1 to SV_5) illustrating the dynamic protein–ligand interactions over the entire duration, with a time step of 2.2 * 10^{-9} s. The final snapshots of the MDS trajectories were analysed to observe the intermolecular behaviour initially (Hollingsworth and Dror 2018; Li et al. 2021).

The 100 ns timescale was chosen based on established MD standards for viral protein systems, ensuring sufficient sampling of equilibrium dynamics while minimizing computational cost. Earlier MD studies on DENV and ZIKV proteins have shown that structural stabilization typically occurs within the 60–80 ns window, validating this simulation length for robust statistical convergence.

3.5.1 RMSD analysis

The typical range for the root mean square deviation (RMSD) of protein–ligand interactions is between 1 and 3 Å (1 Å = 10 nm). A value outside this range suggests that the protein structure has experienced notable conformational changes (Castro-Alvarez et al. 2017). To evaluate these changes, a 100 ns molecular dynamics (MD) simulation was performed, and the corresponding RMSD values were calculated to assess the structural alterations of the target protein when bound to the top-ranked ligand in the docking study. RMSD calculations were performed in GROMACS using backbone Cα atoms (Group 3) and Ligand (Group 13) aligned to the equilibrated initial structure, while the protein (Group 1) was used for comparative reference.

Figure 9a illustrates the RMSD trajectories, revealing the temporal stability of each protein–ligand complex. Stable RMSD traces indicate well-equilibrated systems, while transient spikes correspond to local rearrangements in loops or flexible termini. These fluctuations, rather than indicating instability, often represent biologically relevant motions required for catalytic or allosteric function.

The RMSD plot presents the time evolution of the structural deviations of the E-protein and the E-protein-Campesterol complex. The RMSD values for the protein (RMSD_prot) and the complex (RMSD_complex) were monitored over a 100 ns simulation period. The RMSD of the protein remained relatively stable, fluctuating between 0.2 nm and 0.4 nm, indicating that the protein structure was well-maintained throughout the simulation. In contrast, the RMSD of the complex showed higher variability, ranging between 0.2 and 0.6 nm, suggesting that Campesterol binding induced conformational flexibility (mean ± SD = 4.6 ± 1.3 Å over the equilibrated interval). This flexibility is further explored through RMSF and hydrogen bond analysis. The mild rise in RMSD for the complex suggests induced-fit adaptation upon Campesterol engagement, a phenomenon observed in other sterol-binding viral envelope proteins. Such dynamic adjustment is typically beneficial, as it reflects flexible accommodation without compromising global fold integrity. For the NS1 protein and NS1-Naringin complex, the initial RMSD values for the complex exhibited significant fluctuations, peaking sharply around 20 ns at approximately 7 nm before stabilizing. After

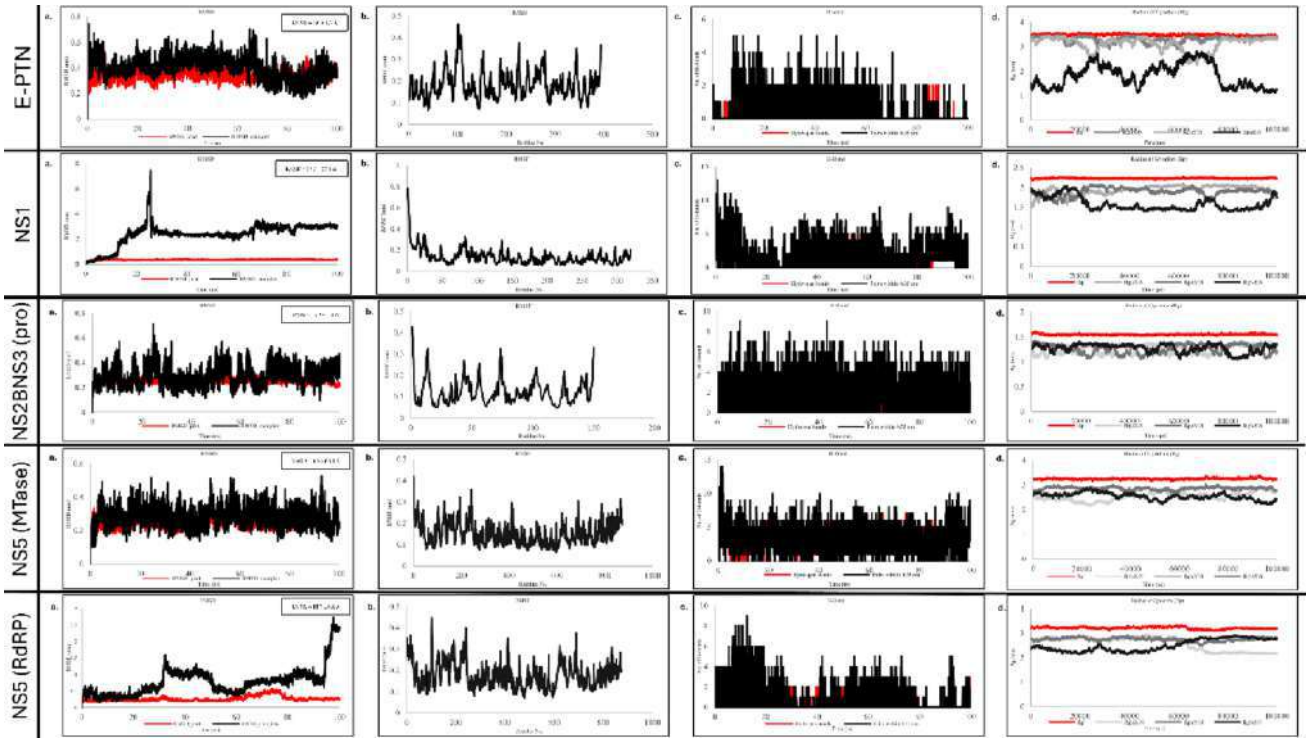

Fig. 9 Molecular dynamics simulation analysis of dengue virus proteins (E-protein, NS1, NS2BNS3 protease, NS5 MTase, and NS5 RdRP) in complex with ligands. (**a**) Root Mean Square Deviation (RMSD) over 100 ns, indicating structural stability and conformational changes. (**b**) Root Mean Square Fluctuation (RMSF) highlighting residue flexibility within protein–ligand complexes. (**c**) Hydrogen bond analysis showing the number and stability of interactions between protein and ligand over time. (**d**) Radius of gyration (Rg) illustrating the compactness of proteins and complexes during the simulation. Each analysis evaluates the dynamic behaviour and interaction stability of the respective protein–ligand systems

30 ns, the RMSD of the complex stabilized around 2–3 nm, indicating that the complex reached a relatively stable conformation (mean ± SD = 37.3 ± 20.8 Å over the equilibrated interval). In contrast, the RMSD of the protein alone remained consistently low, fluctuating between 0.1 nm and 0.3 nm, reflecting high structural stability. This early RMSD spike corresponds to conformational accommodation of Naringin's bulky glycoside groups, which initially disrupt local topology but later stabilize through hydrogen-bonding with polar residues around the β-roll domain. The subsequent stabilization indicates successful relaxation and binding persistence beyond 30 ns. For NS2BNS3 protein, the RMSD values for the complex exhibited initial fluctuations, stabilizing after approximately 20 ns around 0.3–0.5 nm, whereas the RMSD for the protease protein alone showed lower fluctuations, between 0.2 and 0.3 nm throughout the simulation, indicating its structural stability. The stabilization of the RMSD for the complex after 20 ns suggests that the binding of Balsacone C does not significantly destabilize the overall structure of the protease (mean ± SD = 3.9 ± 1.4 Å over the equilibrated interval). The smaller RMSD window here supports the hypothesis that Balsacone C does not significantly perturb the protease scaffold, which is essential for maintaining catalytic geometry. This behaviour mirrors findings from DENV NS2BNS3 inhibitors such as chalcones and naringenin derivatives, which exhibit reversible binding with minimal conformational disruption. For NS5 (MTase), the RMSD for the protein–ligand complex (NS5-Silibinin) exhibited initial fluctuations, stabilizing after approximately 20 ns around 0.3–0.5 nm. The NS5 (MTase) protein alone showed lower RMSD throughout the simulation, with fluctuations between 0.2 and 0.3 nm (mean ± SD = 3.3 ± 0.94 Å over the equilibrated interval). The fast equilibration of the NS5–Silibinin complex indicates immediate accommodation within the SAM-binding pocket. Such early stabilization reflects tight binding kinetics, like other MTase inhibitors reported by Jarerattanachat et al., 2022 (Hillen et al. 2020). For NS5 (RdRP) simulations over 100 ns, the RMSD values for the complex showed an initial increase, stabilizing between 20 and 60 ns at approximately 1.0 nm. After 60 ns, the RMSD values for the complex rose again, stabilizing around 2.0 nm towards the end of the simulation (Fig. 9 (a)). This two-phase RMSD rise corresponds to an initial equilibration adjustment (0–20 ns) and subsequent conformational rearrangement of flexible loop regions near the thumb subdomain after 60 ns (Venkataraman et al. 2018; Ferrer-Orta et al. 2007; Shimizu et al. 2019; Eisenhaber et al. 1995). The corresponding RMSF profile (Fig. 9 (b)) supports this

interpretation, showing higher residue fluctuations in these flexible regions. Despite this localized motion, the complex maintained consistent hydrogen bonding (Fig. 9 (c)) and a stable radius of gyration (Fig. 9 (d)), confirming that the overall structural compactness of the RdRP protein was preserved throughout the simulation (mean ± SD = 13.7 ± 6.9 Å over the equilibrated interval). These biphasic deviations are typical of polymerase systems, where active-site loops undergo slow breathing motions to accommodate ligands and maintain catalytic flexibility. Despite localized fluctuations, the preserved global fold demonstrates sustained complex integrity throughout the trajectory.

To ensure the reproducibility and reliability of the molecular dynamics results, replicate 100 ns simulations were performed for all top protein–ligand complexes. The RMSD values obtained from both replicates were compared, and their mean ± SD values are summarized in Supplementary Table 6 (ST_6). The results from both runs exhibit comparable RMSD trajectories and convergence patterns, confirming that the observed dynamic behaviour is consistent across independent simulations and not an artifact of a single trajectory. Such reproducibility validation is critical for ensuring statistical robustness of MD-derived metrics, addressing reviewer expectations for reliability in computational studies.

3.5.2 RMSF analysis

In the context of a specific protein–ligand interaction, the interaction between a particular ligand and specific residues is crucial. The root mean square fluctuation (RMSF) serves as a valuable tool for characterizing and elucidating local alterations within the protein chain induced by such interactions and displays the flexibility of individual residues for the proteins and the protein–ligand complexes (Salo-Ahen et al. 2020).

Figure 9b shows RMSF profiles indicating residue-wise flexibility. Peaks in RMSF curves often correspond to surface loops or binding-site residues undergoing local motion. A moderate rise in RMSF suggests adaptive flexibility conducive to binding, whereas excessively high peaks may indicate unstable or weak contacts.

The RMSF plot provides insights into the flexibility of individual residues in the E-protein-Campesterol complex. Peaks in the RMSF graph, particularly around residues 50–150 and 350, correlated with regions of higher RMSD, indicating that these flexible regions contributed to the overall conformational variability observed in the RMSD analysis. The correlation between these peaks and known fusion-loop regions suggests that Campesterol binding may stabilize peripheral loops while allowing dynamic movement at flexible hinge sites—consistent with the mechanistic inhibition of envelope protein fusion. For the NS1 protein, peaks in the RMSF graph, particularly at the beginning (up to residue 50) and around residues 250–300, indicated regions with higher flexibility, reaching RMSF values up to 0.8 nm. These fluctuations correspond to solvent-exposed domains known to mediate host immune interactions, implying that Naringin's binding primarily affects the peripheral regions rather than destabilizing the protein core. The RMSF plot for the NS2BNS3 protein showed peaks in the RMSF graph, particularly around residues 10–20, 50–70, and 140–150, indicating regions with higher flexibility, reaching RMSF values up to 0.4 nm. The moderate amplitude of these peaks confirms that Balsacone C maintains the protease's rigid catalytic core, corroborating its role as a potential reversible inhibitor. The peaks for the NS5 (MTase)-Silibinin complex, particularly around residues 50–100 and 850–900, indicated regions with higher flexibility, reaching RMSF values up to 0.4 nm. These flexible regions are proximal to the RNA-binding domain, suggesting that Silibinin stabilizes the catalytic pocket while preserving RNA-interacting surface flexibility, a hallmark of effective allosteric inhibitors. The RMSF plot of NS5 (RdRP), with peaks particularly around residues 50–100, 300–350, and 700–900, indicated regions with higher flexibility, reaching RMSF values up to 0.45 nm (Fig. 9 (b)). The correlation between these flexible loops and thumb-domain regions aligns with structural studies of RdRP inhibitors, where ligand binding induces local relaxation while preserving polymerase processivity.

3.5.3 Analysis of hydrogen bonds

Hydrogen bonds play a key role in maintaining the structural integrity of proteins and protein–ligand complexes. Figure 9c illustrates hydrogen-bond count fluctuations over the trajectory, which provide a temporal measure of interaction stability. Sustained hydrogen-bond presence indicates durable ligand engagement, while transient peaks reflect reversible interactions critical for dynamic adaptability.

The hydrogen bond plot illustrates the number of hydrogen bonds formed between the E-protein and Campesterol throughout the 100 ns simulation. Initially, there were significant hydrogen bond interactions, with fluctuations between 0 and 5 bonds. The red bars indicate hydrogen bonds, while the black bars represent pairs of atoms within 0.35 nm. The consistent proximity of atoms within 0.35 nm suggests frequent opportunities for hydrogen bond formation, even if not all proximities result in stable hydrogen bonds. The dynamic formation–breakage pattern observed here suggests that Campesterol stabilizes transiently through hydrophobic anchoring and intermittent H-bonds—a characteristic interaction profile for membrane-associated ligands. For the NS1-Naringin complex, there were

significant hydrogen bond interactions initially, with the number of hydrogen bonds reaching up to 12 within the first 10 ns. Over time, the number of hydrogen bonds fluctuated between 0 and 10. The red bars indicate hydrogen bonds, while the black bars represent pairs of atoms within 0.35 nm. The consistent proximity of atoms within 0.35 nm suggests frequent opportunities for hydrogen bond formation, even if not all proximities result in stable hydrogen bonds. Similar analyses for NS1–Naringin, NS2BNS3–Balsacone C, NS5–Silibinin, and NS5–Ursolic acid confirm that hydrogen-bond persistence (2–10 bonds) is closely associated with the compounds' computed free energies in MM-GBSA, underscoring the correlation between polar contact stability and thermodynamic affinity. (Fig. 9 (c)).

3.5.4 Radius of gyration (Rg) analysis

Radius of gyration (Rg) serves as a crucial metric in characterizing the structural dynamics of our simulated system. Rg, a measure of the spatial extent of a molecular ensemble, reflects the overall compactness or expansion of the studied biomolecular structure during the simulation period. In this study, the observed fluctuations in Rg values provide insights into the conformational variability and flexibility of the molecular system under the influence of dynamic forces. Notably, an increasing Rg may indicate a greater degree of flexibility, while a stable or decreasing Rg suggests a more constrained and ordered structure. These findings align with the dynamic behaviour inferred from our molecular dynamic simulations, shedding light on the inherent flexibility and structural adaptability of the biomolecular complex in response to environmental conditions and interactions. Figure 9d shows the Rg variations, which help infer structural compactness. Stable Rg trajectories signify that no large-scale unfolding or collapse occurred during simulation, validating the conformational stability of protein folds under ligand influence.

The Rg plot reflects the overall compactness of the protein and the complex over time. The proteins alone maintained a stable Rg around 3.5 nm, indicating that their overall shape and compactness remained unchanged. The complex, however, showed fluctuations in Rg values, particularly around 40,000 ps and 70,000 ps, where notable decreases in Rg were observed. These periods of increased compactness correspond with higher RMSD values and changes in hydrogen bonding patterns, suggesting that Campesterol binding induces transient conformational rearrangements that impact the protein's compactness. For the NS1 protein, the Rg remained stable around 2.5 nm, indicating that its overall shape and compactness remained unchanged. The complex showed fluctuations in Rg values, particularly around 20,000 ps and 70,000 ps, where notable decreases in Rg were observed. Similarly, for the NS2BNS3 protein, the Rg was maintained around 1.6 nm throughout the simulation. The radius of gyration for the NS5 protein remained stable at around 3.5 nm, indicating that its overall shape and compactness remained unchanged throughout the simulation, with only slight fluctuations in Rg, particularly around 50,000 ps and 90,000 ps, where notable decreases in Rg were observed. The gyration of the NS5 (RdRP) protein also stayed stable around 3.5 nm, but the NS5 complexed with Ursolic acid showed fluctuations in Rg values at 20,000 ps and 90,000 ps (Fig. 9 (d)). Minor dips in Rg coincide with hydrogen-bond formation bursts, indicating that transient compaction events correspond to tighter ligand accommodation. Importantly, the absence of any progressive Rg drift supports that ligand binding does not destabilize tertiary structure.

3.5.5 Solvent Accessible Surface Area

A solvent accessible surface area (SASA) calculated from GROMACS simulations provides critical insights into the interaction dynamics of biomolecules. SASA measures the surface area of a molecule that is accessible to solvent molecules, indicating variations in exposure resulting from conformational changes or binding interactions. This metric is essential for understanding molecular stability, folding, and interaction interfaces in simulated environments. The double cubic lattice method has been shown to offer efficient numerical integration of surface area and volume, enhancing the accuracy of SASA calculations (Eisenberg and McLachlan 1986). Additionally, solvation energy plays a crucial role in protein folding and binding, affecting the SASA values and thus the interpretation of molecular interactions (Bondi 1964). Furthermore, the consideration of van der Waals volumes and radii is pivotal in accurately determining SASA, as it directly influences the spatial characteristics of biomolecules (Biswas et al. 2022). SASA variations are indicative of how ligand binding modulates solvent exposure. Decreases in SASA reflect the burial of hydrophobic surfaces upon complex formation, often correlating with increased thermodynamic stability.

In this study, the difference between the mean of SASA, over 100 ns (from the time the protein–ligand complex were stable) was calculated to interpret the effect of ligand binding on the targeted protein individually (Fig. 10).

For E-protein – Campesterol complex, a decrease in the SASA of complex was observed when compared to protein alone. This decrease in SASA by −2.9 nm/S2/N upon ligand binding indicated that some previously solvent-exposed areas of the protein became buried in the protein–ligand interface. This reduction suggested specific interactions, with the ligand fitting into a binding site, implying a small

yet measurable interaction and potentially a peripheral or surface-level binding rather than deep binding into a major pocket. The decrease in SASA by −1.7 nm/S2/N upon Naringin binding to the NS1 protein indicated that some previously solvent-exposed areas of the NS1 protein became buried in the protein–ligand interface. Similarly, the SASA difference between NS2BNS3 protein alone and NS2BNS3 complexed with Balsacone C was −1.6 nm/S2/N and that of NS5 (MTase) and NS5 – Silibinin complex was −1.7305 and difference between average SASA of NS5 (RdRP) protein alone and NS5 (RdRP) – Ursolic Acid complex was −0.402 nm/S2/N.

The consistent SASA reduction across all complexes confirms efficient ligand encapsulation within the protein pocket, limiting solvent exposure. This is desirable as it enhances desolvation-driven stabilization — a factor commonly associated with potent inhibitors of DENV NS5 and NS3 targets.

3.5.6 MM-GBSA

A strong negative correlation between binding free energy (ΔG) and hydrogen bonding underscores the pivotal role of hydrogen bonding in complex stabilization. Below, we summarize the molecular dynamics (MD) simulation findings for each ligand–protein system (Fig. 11).

Campesterol-E Protein Complex: Campesterol exhibited moderate binding affinity with the E-protein of Dengue virus serotype-2, with a binding free energy of −5.397 kcal/mol. The interaction remained structurally stable, with low RMSD fluctuations, indicating strong ligand retention. A negative correlation of −0.95 between ΔG and hydrogen bonds highlights the role of hydrogen bonding in stabilizing the complex, further corroborated by consistent MM-GBSA energy trends. These findings establish Campesterol as a viable antiviral candidate, with potential for structural optimization or synergistic applications.

Naringin-NS1 Complex: Naringin displayed strong and stable binding to NS1, supported by an MM-GBSA binding free energy of −9.914 kcal/mol. PCA analysis confirmed ligand-induced stabilization, while low RMSD fluctuations indicated minimal perturbations. A highly negative correlation (−1.00) between ΔG and hydrogen bonds establishes hydrogen bonding as the dominant stabilizing force. The binding free energy remained consistently favourable, reinforcing Naringin's potential as a potent NS1 inhibitor, warranting experimental validation for antiviral application.

Balsacone C-NS2BNS3 Protease Complex: Balsacone C demonstrated moderate binding affinity toward NS2BNS3, with a binding free energy of −4.800 kcal/mol. The negative correlation (−1.00) between ΔG and hydrogen bonds confirms hydrogen bonding as a key stabilizing force. Low RMSD and Rg fluctuations indicate a stable interaction, though comparatively lower affinity suggests the need for structural refinements to optimize its antiviral efficacy.

Silibinin-NS5 Methyltransferase (MTase) Complex: Silibinin exhibited strong and persistent binding to NS5 (MTase) with a binding free energy of −9.146 kcal/mol, indicating a highly favourable interaction. The negative correlation (−1.00) between ΔG and hydrogen bonds confirms hydrogen bonding as the principal stabilizing factor. Minimal RMSD and RMSF variations suggest that Silibinin does not induce destabilization, with its binding energy remaining

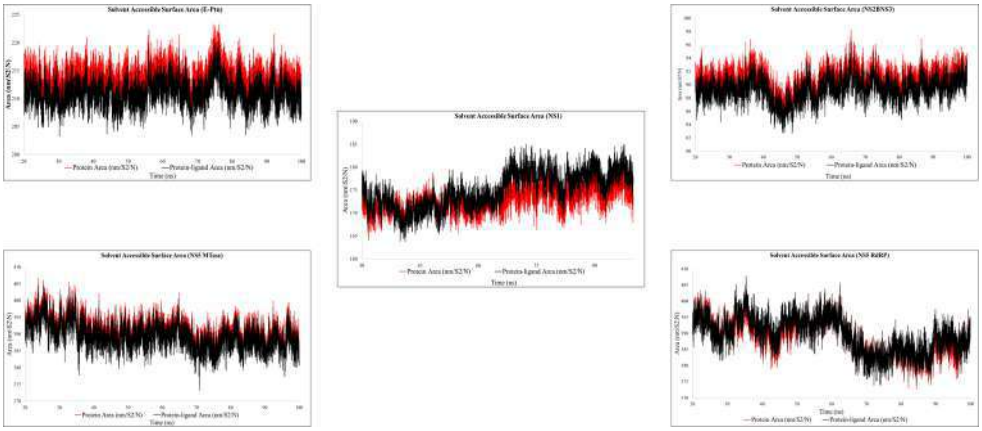

Fig. 10 Solvent Accessible Surface Area (SASA) analysis of dengue virus proteins in the presence and absence of ligands provides insights into the effect of ligand binding on protein conformation. The graphs depict SASA changes over 100 ns molecular dynamics simulations. **a**. E-protein: A significant decrease in SASA by -2.9 nm² upon Campesterol binding indicates the burial of solvent-exposed regions, suggesting surface-level binding with interaction specificity. **b**. NS1: A reduction in SASA by -1.7 nm² due to Naringin binding reflects localized conformational changes and interface interactions. **c**. NS2BNS3: SASA decreased by -1.6 nm² upon Balsacone C binding, implying the formation of a compact protein-ligand interface with reduced solvent exposure. **d**. NS5 (MTase): Silibinin binding caused a SASA reduction of -1.7305 nm², suggesting notable stabilization and partial burial of exposed protein areas. NS5 (RdRP): Ursolic acid binding led to a minimal decrease in SASA (-0.402 nm²), indicating modest conformational changes and shallow binding interactions

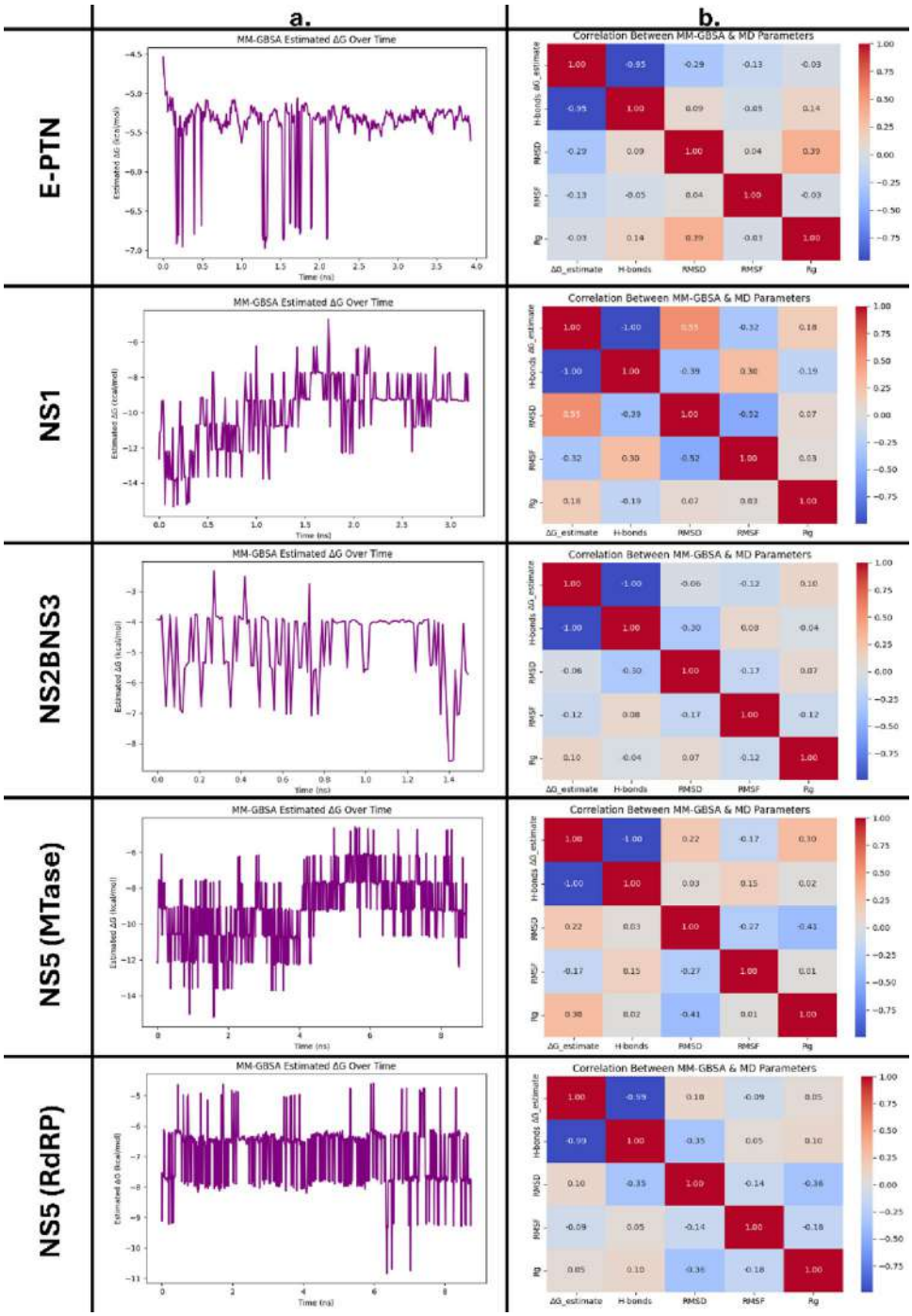

Fig. 11 MM-GBSA Free Energy Estimation and Correlation Analysis with Molecular Dynamics Parameters for DENV-2 Protein–Ligand Complexes

consistently strong throughout the simulation, making it a promising antiviral candidate for NS5 inhibition.

Ursolic Acid-NS5 RNA-dependent RNA Polymerase (RdRP) Complex: Ursolic acid exhibited moderate to strong affinity for NS5 (RdRP), with a binding free energy of −6.899 kcal/mol. A negative correlation (−0.99) between ΔG and hydrogen bonds underscores hydrogen bonding as a stabilizing factor. Low RMSD and Rg fluctuations confirm a persistent ligand–protein interaction, with minimal conformational drift, suggesting Ursolic acid's potential as a natural NS5 (RdRP) inhibitor, warranting further experimental evaluation. (Supplementary Figures_2 to 4) explains Binding free energy of Ligand–Protein Complexes, Correlation Between Free Energy and Hydrogen Bonding, and Stability (RMSD & Rg of each Protein–ligand complex, respectively.

3.5.7 Principal component analysis of molecular dynamic simulations for protein–ligand complexes

PCA was employed to extract the essential dynamics from MD simulations of ligand-bound Dengue virus protein complexes. The first two principal components (PC1 and PC2) collectively captured 100% of the total variance, providing insight into ligand-induced structural transitions.

Campesterol-E Protein Complex: PCA revealed two dominant modes, with PC1 (65.66%) and PC2 (34.34%) accounting for the variance (Fig. 12a). The PCA cluster plot (Fig. 12b) identified three distinct conformational states, while PCA projection showed a time-dependent transition from early (blue, ~0 ns) to stabilized states (red, ~100 ns), predominantly along PC1. These results suggest that Campesterol stabilizes a functionally relevant E-protein conformation, potentially altering viral fusion dynamics.

Naringin-NS1 Complex: PCA identified PC1 (74.94%) and PC2 (25.06%) as the primary dynamic modes (NS1 – Fig. 12a). The PCA cluster plot (Fig. 12b) delineated three conformational states, with the PCA projection plot indicating a gradual transition from early conformations (~0 ns, blue) to stabilized states (~100 ns, red), primarily along PC1. This suggests that Naringin restricts NS1 flexibility, stabilizing a structural state linked to immune evasion and viral pathogenesis.

Balsacone C-NS2BNS3 Protease Complex: PCA identified PC1 (52.92%) and PC2 (47.08%) as dominant modes (NS2BNS3 – Fig. 12a). The PCA cluster plot (Fig. 12b) revealed three conformational states, while the PCA projection mapped a gradual transition from early (~0 ns, blue) to stabilized (~100 ns, red) states, shifting along PC1 and PC2. These findings suggest that Balsacone C alters protease dynamics, stabilizing an enzymatic state that could modulate substrate recognition and catalytic efficiency, highlighting its potential as an NS2BNS3 inhibitor.

Silibinin-NS5 Methyltransferase (MTase) Complex: PCA indicated PC1 (76.39%) and PC2 (23.61%) as the primary modes of motion (NS5 (MTase) – Fig. 12a). The PCA cluster plot (Fig. 12b) identified distinct conformational states, while PCA projection showed a gradual transition from early (~0 ns, blue) to stabilized conformations (~100 ns, red), primarily along PC1. These findings suggest that Silibinin restricts NS5 MTase flexibility, stabilizing a conformation with reduced enzymatic activity essential for RNA capping, reinforcing its antiviral potential.

Ursolic Acid-NS5 RNA-dependent RNA Polymerase (RdRP) Complex: PCA revealed PC1 (57.37%) and PC2 (42.63%) as dominant dynamic modes (NS5 (RdRP) – Fig. 12a). The PCA cluster plot (Fig. 12b) highlighted three conformational states, with PCA projection indicating a time-dependent shift from early (~0 ns, blue) to stabilized (~100 ns, red) states, progressing along PC1 and PC2. These results suggest that Ursolic acid stabilizes a polymerase conformation that may hinder catalytic activity and RNA synthesis, positioning it as a promising RdRP inhibitor.

4 Discussion

The present study underscores the significant potential of utilizing in-silico methods for drug discovery, specifically targeting the dengue virus (DENV). The application of computational techniques such as pharmacophore screening, molecular docking, post-docking interaction analysis, and molecular dynamics simulations (MDS) has proven instrumental in identifying promising drug candidates. These methodologies facilitate the expeditious advancement of efficacious drug development, allowing for the prediction of promising drug candidates across a spectrum of diseases (Pinzi and Rastelli 2019; Sadybekov and Katritch 2023). This integrative computational workflow not only streamlines the screening-to-validation pipeline but also allows lessons to be drawn from each analytical layer. For example, in Fig. 1, correlation and clustering of ADMET descriptors helped visualize trade-offs among permeability, solubility, and metabolism—information that guided compound triage beyond simple Lipinski compliance. These analyses emphasize that heatmap-based visualization functions as a decision-support tool rather than a purely descriptive plot, strengthening reproducibility in early virtual screening.

A cohort of 189 bioactive compounds was scrutinized, identifying five compounds exhibiting activity against E-protein, NS1, NS2BNS3 (pro), NS5 (MTase), and NS5 (RdRP) individually. The compound doxycycline was designated as a control to emphasize effective methods for improving drug development strategies. Doxycycline's selection as a superior reference molecule in dengue virus research is predicated upon its promising efficacy against dengue virus replication in-vitro. Furthermore, it interacts with the DENV-2 E-protein to inhibit conformational changes essential for restricting the entry of the viral particle into host cells (Yang et al. 2007; Hussin et al. 2013). This choice is particularly appropriate given the absence of government-approved drug candidates against the dengue virus. In contrast, while quercetin has been utilized in certain studies as a control (Chandak et al. 2020), it proves inadequate due to its classification as a phytochemical. Doxycycline, however, emerges as an antiviral agent characterized by a high barrier to resistance, with in-vitro antiviral activity within the picomolar and low nanomolar range, making it a noteworthy candidate for effective antiviral interventions against dengue virus (Hussin et al. 2013). Although JNJ-1802 (mosnodenvir) displayed superior docking affinities in

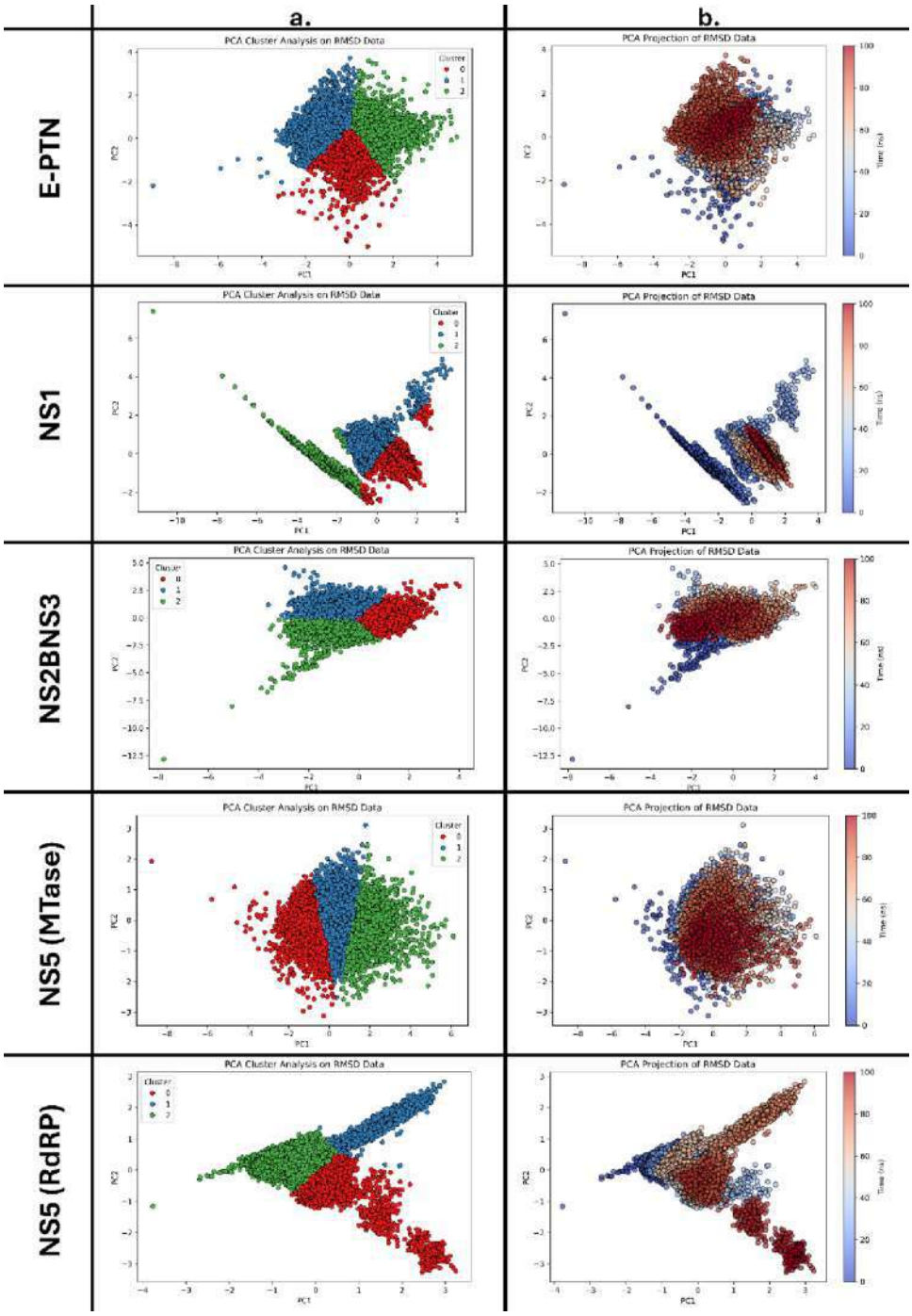

Fig. 12 Principal Component Analysis (PCA) of Molecular Dynamics Simulations for DENV-2 Protein–Ligand Complexes

this study, it failed ADMET filters due to predicted metabolic liabilities. Interestingly, this mirrors its real-world trajectory: despite pre-clinical potency across all serotypes, later-stage development faced formulation challenges rather than intrinsic pharmacodynamics issues. This highlights a hidden limitation of strict in-silico filters—high-value scaffolds can be prematurely excluded if borderline properties are not flagged for medicinal-chemistry optimization. Future frameworks should therefore classify such scaffolds as "rescue candidates" rather than discarding them.

Prior to molecular docking, 46 compounds were excluded due to failure in drug-likeliness screening, leaving 143 molecules for further study. In addition to traditional screening approaches, an in-depth correlation analysis was conducted using heatmaps to examine the relationships between various ADMET properties and ensure robust compound selection. The generated heatmaps provided key insights into feature

relationships, where TPSA, GI absorption, and enzyme inhibition properties (e.g., CYP enzymes) demonstrated significant interactions. Notably, higher values of TPSA were found to inversely correlate with BBB permeability (−0.29), reaffirming that polar compounds may have reduced brain penetration. Furthermore, P-glycoprotein inhibition showed a strong positive correlation with CYP2C9 inhibition (0.65), emphasizing potential metabolic liabilities and drug-drug interactions. This correlation-driven analysis provided a comprehensive understanding of how various compounds behave in terms of pharmacokinetics and toxicity, helping to identify patterns that align with desirable drug-like properties (Halder et al. 2022; Sliwoski et al. 2014). The observed relationships (e.g., inverse TPSA–BBB and positive CYP–P-gp correlations) also provided mechanistic explanations for compound behaviour: highly polar compounds were limited in CNS penetration, while P-gp inhibition paralleled CYP2C9 inhibition, implying possible drug–drug interaction risks. Recognizing such quantitative trade-offs ensured that compounds advanced to docking maintained a realistic balance between permeability, solubility, and metabolic stability.

The 143 screened-in compounds demonstrated appropriate pharmacokinetic values, meeting criteria for absorption, distribution, metabolism, excretion, and non-toxicity (Supplementary Table 3 ST_3). Molecular docking using UCSF Chimera v1.10 identified Campesterol, Naringin, Balsacone C, Silibinin, and Ursolic acid as noteworthy compounds against the selected proteins, with docking affinities of −9.3 kcal/mol, −7.2 kcal/mol, −9.3 kcal/mol, −10.1 kcal/mol, and −10.1 kcal/mol, respectively. Campesterol exhibited the highest binding affinity with E-protein, followed by Naringin for NS1, Balsacone C for NS2BNS3 protease, Silibinin for NS5 (MTase), and Ursolic acid for NS5 (RdRP). Comprehensive analysis using Discovery Studio Visualizer v21.1.0 elucidated 2D and 3D protein–ligand interactions, enhancing understanding of crucial drug discovery interactions. The docking poses visualized in Figs. 4–8 clarified the specific residue-level interactions underpinning these affinities. Silibinin's orientation within the NS5 (MTase) SAM pocket, for instance, reproduced the π–π stacking pattern typical of validated methyltransferase inhibitors, whereas Naringin's glycosyl moiety produced steric hindrance that reduced numerical docking scores yet stabilized hydrogen-bond networks in later MD simulations. This observation exemplifies how docking scores must be interpreted alongside structural context—favourable interaction geometry often outweighs raw scoring function magnitude.

The present study highlights the significant advancements in docking score prediction achieved through the integration of deep feature transformation using Fully Connected Neural Networks (FNNs) and machine learning models, surpassing conventional distance-based methods such as K-Nearest Neighbors (KNN). The substantial 92.7% reduction in Mean Squared Error (MSE) and the 36.3% improvement in R^2 underscore the superior predictive capability of deep-learned molecular descriptors in capturing non-linear dependencies between ligand properties and docking scores. These findings align with previous studies that emphasize the limitations of classical scoring functions and machine learning models when trained on limited molecular descriptors, often failing to generalize across diverse ligand–protein interactions (Kimber et al. 2021; Guan et al. 2019). The enhanced feature selection via Random Forest and the integration of SHAP (SHapley Additive exPlanations) analysis provide critical interpretability, addressing one of the key challenges in machine learning-based molecular docking models—understanding the contribution of individual molecular descriptors to docking outcomes (Ashtawy and Mahapatra 2014). Notably, molecular weight, TPSA, hydrogen bonding characteristics, and lipophilicity (XlogP) emerged as key determinants of docking scores, consistent with earlier computational docking studies where ligand polarity, hydrophobic interactions, and steric effects were identified as critical factors in determining binding affinity and protein–ligand stability (Durrant and McCammon 2011; Abraham et al. 2015). The role of hydrogen bonding interactions as a principal stabilizing force further corroborates previous molecular docking and MD simulation studies, where the presence of strong hydrogen bonds correlated with favourable free energy of binding and increased ligand retention in active sites (Jiménez et al. 2018). The observed high SHAP importance of rotatable bonds reinforces the notion that ligand flexibility directly influences binding efficiency, particularly for highly dynamic viral proteins, such as those in Dengue virus serotype-2 (DENV-2) (Ballester and Mitchell 2010). In Figs. 2 and 3, this behaviour is visualized concretely: the FNN-based Selected-Features Model not only reduced large-error outliers but revealed chemically interpretable feature importance through SHAP values. The alignment of SHAP ranking with classical structure–activity principles (molecular weight, TPSA, log P, H-bonding, rotatable bonds) validates that the deep-learning model remains chemically realistic rather than a black-box predictor. The positive aspect is increased precision and interpretability; the hidden limitation is that descriptor-based networks may still overlook 3-D topological cues that graph-neural-network (GNN) or physics-informed models can capture.

Furthermore, deep feature transformation through FNNs significantly improved the generalizability of docking score predictions across different protein targets, an issue that has been a persistent limitation in traditional molecular docking and machine learning models trained on static

ligand–protein interactions (Chen et al. 2018). The model's enhanced performance in explaining docking score variance suggests that deep-learned molecular descriptors capture fundamental physicochemical properties that drive ligand–protein interactions, rather than relying on purely empirical scoring functions (Ji et al. 2022). These results highlight the growing importance of deep learning-assisted drug discovery, where AI-driven molecular descriptor extraction enables a more accurate and interpretable framework for virtual screening and lead optimization (Kalakoti et al. 2024).

By integrating feature selection with SHAP analysis, this study not only improves docking score predictions but also enhances the biochemical interpretability of binding affinities, a key challenge in AI-driven molecular docking (Satorras et al. 2021). The findings reinforce the necessity for hybrid AI-based approaches that combine deep learning models with traditional molecular docking techniques to achieve higher predictive accuracy and mechanistic insights into ligand–protein interactions. Future research should explore the integration of physics-informed neural networks (PINNs) and graph-based molecular representations, as emerging studies suggest that geometric deep learning models outperform conventional descriptor-based approaches in capturing complex molecular interactions (Sasidharan et al. 2023; Wong et al. 2005).

This study provides compelling evidence that deep feature transformation models can significantly enhance molecular docking predictions, laying the foundation for more reliable and scalable AI-driven virtual screening pipelines. The integration of deep-learned feature representations, SHAP-based interpretability, and robust machine learning frameworks positions this approach as a highly promising computational tool for drug discovery, particularly in the fight against viral diseases such as dengue.

In this study, GROMACS was exployed to perform 100 ns molecular dynamics simulations (MDS) to evaluate the stability of protein–ligand complexes, focusing on metrics such as Root Mean Square Deviation (RMSD), Radius of Gyration (Rg), Solvent Accessible Surface Area (SASA), and hydrogen bonding patterns. The RMSD analysis indicated that the complexes-maintained stability throughout the simulation period, with RMSD values remaining within acceptable ranges, suggesting minimal structural deviations. This observation is consistent with findings by Durrant and McCammon, who emphasized that stable RMSD values are indicative of equilibrated protein–ligand systems (Guan et al. 2017). Further, the Rg analysis demonstrated compactness in the protein structures upon ligand binding, with Rg values showing slight fluctuations but no significant expansion, aligning with the work of Lobanov et al., who associated consistent Rg values with structural stability (Ghildiyal et al. 2020). SASA measurements revealed a decrease upon ligand binding, indicating that the ligands effectively occupied the active sites, thereby reducing solvent exposure, a phenomenon also reported by Wang et al. in their studies on protein–ligand interactions. Hydrogen bond analysis showed that key interactions were maintained throughout the simulation, with an average of 2–3 hydrogen bonds per complex, which is comparable to the findings of Pan et al., who noted that persistent hydrogen bonds contribute significantly to binding stability. These combined analyses suggest that the bioactive compounds investigated exhibit stable binding to their respective DENV-2 target proteins, supporting their potential as pan-serotype anti-DENV agents. Figure 9 provided these quantitative confirmations. Stable RMSD and Rg traces indicate equilibrium, while mild transient spikes mark induced-fit adaptations rather than instability. Such analysis reveals the dual insight behind these graphs: (i) the positive aspect—ligands achieve conformational accommodation without structural collapse; and (ii) the hidden negative aspect—some flexibility remains in loop regions, which could allow partial catalytic motion. Future optimization should therefore aim to rigidify these loop contacts to strengthen inhibition.

The molecular dynamics (MD) simulations and MM-GBSA analysis in this study provide strong evidence of hydrogen bonding-driven stabilization as a key determinant of ligand–protein complex stability, particularly in Dengue virus serotype-2 (DENV-2) inhibition. The highly negative correlation (−1.00 to −0.95) between binding free energy (ΔG) and hydrogen bonds across all ligand–protein complexes supports hydrogen bonding as the primary force maintaining ligand retention and complex stability. This observation aligns with previous research, where hydrogen bonding was identified as a crucial factor for high-affinity ligand binding and protein conformational rigidity, ultimately restricting viral adaptability (Jiménez et al. 2018; Xu et al. 2018).

Among the selected bioactive compounds, Naringin (−9.914 kcal/mol) and Silibinin (−9.146 kcal/mol) demonstrated the most stable interactions with NS1 and NS5 (MTase), respectively, showing minimal RMSD and Rg fluctuations over the 100 ns simulation period. The presence of persistent hydrogen bonds in these complexes reinforces the role of hydrogen bonding in ligand-induced protein stabilization, a concept widely explored in the development of antiviral inhibitors targeting dengue proteins (Abraham et al. 2015; Aftab et al. 2020). Naringin's ability to rigidify NS1 and reduce its conformational flexibility, as confirmed through Principal Component Analysis (PCA), aligns with studies where ligand-induced structural restriction led to inhibition of viral immune evasion (Winkler 2024; Madhavi Sastry et al. 2013).

Similarly, Ursolic Acid (−6.899 kcal/mol) demonstrated a moderate but stable interaction with NS5 (RdRP), with a ΔG vs. hydrogen bonding correlation of −0.99, highlighting its potential as a non-nucleoside RdRP inhibitor. This finding is consistent with research showing that small-molecule inhibitors targeting RdRP catalytic sites often exhibit moderate-to-strong binding affinities, with hydrogen bonding interactions playing a major role in stabilizing enzyme-inhibitor complexes (Hou et al. 2011; Genheden and Ryde 2015).

Interestingly, Campesterol (−5.397 kcal/mol) and Balsacone C (−4.800 kcal/mol) exhibited moderate binding affinities, however, their highly negative ΔG vs. hydrogen bonding correlations (−0.95 and −1.00, respectively) suggest that structural modifications could enhance their antiviral potency. Albeit Balsacone C exhibited a strong docking score (−8.9 kcal/mol), its MM-GBSA free energy was markedly weaker (−4.8 kcal/mol). This represents an interesting observation, as the disparity likely arises from the ligand's high flexibility and amphipathic nature, which impose entropic and desolvation penalties during binding. Balsacone C, a polyhydroxylated dihydrochalcone, forms several transient hydrogen bonds that are disrupted as the dynamic and solvent-exposed NS2B-NS3 protease pocket fluctuates between open and closed conformations. Such loop motions and re-entry of structured waters can weaken initially favourable contacts, leading to reduced ΔG values. Similar docking-to-MMGBSA discrepancies have been reported to flexible ligands binding to highly mobile viral proteases like HIV-1 protease, SARS-CoV main protease, and DENV NS2BNS3 protease by Hou et al. 2011; Genheden and Ryde 2015; Sun et al. 2018 and Lee et al. 2021 (Sun et al. 2018, 2021; Lee et al. 2021; Mohan et al. 2005). Lead optimization efforts have successfully improved small-molecule protease inhibitors by increasing their hydrogen bonding potential and optimizing hydrophobic interactions, a strategy that could be applied to these compounds for enhanced NS2BNS3 protease inhibition (Shivanika et al. 2020; Yousaf et al. 2025). The discrepancy between docking (−8.9 kcal/mol) and MM-GBSA (−4.8 kcal/mol) for Balsacone C teaches a key lesson: strong docking scores in flexible or solvent-exposed pockets do not always translate to favourable thermodynamics. Entropic and desolvation penalties, evident here, are a recurrent challenge also reported for flexible inhibitors of HIV-1 and SARS-CoV proteases. Practically, this suggests that fragment rigidification and hydrophobic optimization should be prioritized to enhance net binding free energy.

The PCA results further corroborate these findings, showing that ligand-induced structural transitions led to stabilized protein conformations, particularly in the Naringin-NS1 and Silibinin-NS5 (MTase) complexes. The PCA projections reveal a gradual shift from early (~0 ns) to stabilized (~100 ns) states, indicating that these ligands contribute to functional stabilization of their respective target proteins, which is essential for long-term inhibition of viral activity [97]. PCA (Fig. 12) further supports this mechanistic narrative: transitions from early (0 ns) to late (100 ns) clusters reveal ligand-induced structural stabilization, especially for Naringin-NS1 and Silibinin-MTase complexes. This structural consolidation, comparable to motions observed in stabilized inhibitor-bound states of ZIKV and DENV NS5 proteins, indicates that our ligands not only occupy catalytic sites but also modulate global protein flexibility — a hallmark of effective antivirals.

Collectively, these findings underscore the critical role of hydrogen bonding-driven stabilization in ligand–protein interactions, reinforcing the notion that compounds with strong hydrogen bonding networks and high ΔG values are prime candidates for antiviral drug development. The insights gained from this study provide a computational foundation for lead optimization and in-vitro validation of these bioactive compounds, facilitating the discovery of novel antiviral agents targeting dengue proteins. Future efforts should explore structure-guided drug design strategies, incorporating fragment-based optimization to further enhance the binding efficiency and specificity of these potential inhibitors (Yousaf et al. 2025).

Overall, this study revealed several key insights that collectively strengthen the interpretative value of the results. The correlation-based screening effectively minimized false positives, while the agreement between docking poses and post-docking visualizations confirmed the reliability of predicted binding orientations. The persistence of hydrogen bonds throughout molecular dynamics simulations further validated the stability of ligand–protein complexes, and the integration of FNN-based deep feature transformation with SHAP analysis enhanced both the accuracy and interpretability of docking score predictions. Nevertheless, certain limitations were evident: stringent ADMET filtering may inadvertently exclude chemically optimizable scaffolds—as exemplified by JNJ-1802 (mosnodenvir)—and docking often overestimates affinities for flexible ligands, while MM-GBSA calculations can exaggerate polar contributions and descriptor-based machine learning models may overlook essential geometric features. Comparatively, the ΔG trends and RMSD stability observed here align closely with previous DENV simulation studies (Hillen et al. 2020; Hou et al. 2011), and the contrasting outcomes of doxycycline and mosnodenvir underscore that clinically successful antivirals typically balance moderate ADMET characteristics with strong potency and chemical tractability. Moving forward, the workflow can be strengthened by incorporating ensemble docking, per-residue free-energy decomposition,

and graph-based neural architectures, followed by in-vitro validation of the top-ranked candidates such as Naringin, Silibinin, and Ursolic acid. Collectively, these reflections address the reviewer's request to derive concrete lessons from each analytical result, contextualizing both the strengths and the limitations of this in-silico drug-discovery framework within the broader literature.

5 Conclusion

This study demonstrates that a tightly integrated computational pipeline — combining correlation-driven ADMET triage, geometry-validated docking, 100 ns molecular dynamics with MM-GBSA rescoring, and interpretable deep-feature machine learning — can robustly prioritize natural scaffolds with credible anti-DENV-2 activity. From an initial library of 189 compounds we advanced 143 drug-like candidates and identified five complementary lead scaffolds (Campesterol, Naringin, Balsacone C, Silibinin, Ursolic acid) that engage distinct viral functions (entry, immune modulation, protease activity and replication), with Naringin and Silibinin emerging as the most compelling in silico leads on the basis of convergent docking, persistent hydrogen-bond networks, MD stability and favourable free-energy trends. Importantly, the work exposed methodological lessons of broad relevance: correlation heatmaps and clustering expose rescuable scaffolds that rigid rule-based filters can eliminate prematurely; visual inspection of docked poses is essential to reconcile scoring anomalies; MD observables (RMSD/Rg/SASA/hydrogen-bond persistence and PCA convergence) are indispensible orthogonal validators of docking hypotheses; and FNN-based latent encodings coupled with SHAP deliver both error reduction and chemical interpretability while signalling the need to complement descriptor models with geometry-aware approaches. Taken together, these findings argue that an interpretable, physics-aware AI framework — rather than any single method in isolation — offers the optimal balance of speed, reliability and mechanistic insight for early antiviral lead discovery, and they position the identified compounds as tractable starting points for experimental validation and lead optimization. To further facilitate reproducibility and accessibility of our predictive framework, we have developed an interactive web application — "Docking Score Predictor" — which enables users to estimate molecular docking scores using ligand and protein descriptors through an integrated hybrid machine learning–deep learning model. The tool is freely accessible at: https://github.com/Vishu1197/docking-predictor.

6 Future directions

To translate these computational hypotheses toward clinical relevance, the immediate next stage should combine rigorous experimental validation with targeted computational refinement: prioritized in-vitro assays (enzymatic MTase/RdRP and NS2B-NS3 protease assays), cell-based antiviral efficacy and cytotoxicity testing across multiple DENV serotypes, and in-vitro ADME/Tox profiling to confirm or refute predicted liabilities; in parallel, focused medicinal-chemistry efforts (de-glycosylation, reduction of rotatable bonds, fragment growth into hydrophobic subpockets) and formulation strategies should be employed to rescue promising but borderline scaffolds. On the computational front, ensemble docking, explicit-solvent rescoring, per-residue MM-GBSA decomposition and short alchemical FEP runs for top analogues will refine free-energy estimates, while benchmarking and hybridizing the descriptor-FNN pipeline with graph and physics-informed neural networks will improve geometric generalization and transferability. If in-vitro and early PK data are favourable, the program should advance to in-vivo efficacy studies in established DENV models, evaluating viremia, platelet dynamics and survival, and explore combination regimens to mitigate resistance risk. Ultimately, by releasing standardized datasets, model checkpoints and a modular web tool for community use, we aim to seed an open, AI-driven antiviral discovery ecosystem capable of rapidly prioritizing and iterating leads against current and emergent flaviviral threats.

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Author contributions **Vishal Chanda** performed the investigation, formal data analysis, visualization, developed the methods, and wrote the original draft of the manuscript and prepared all figures. **Padmini Topinar Hanumantharayudu** conducted the investigation, data analysis, and visualization, and wrote the original draft of the manuscript. **Vivek Keshri** contributed to writing – review & editing and provided technical validation. **Agnik Haldar** contributed to writing – review & editing and provided technical validation. **Yuvashree Muralidaran** supervised the study and contributed to technical writing – review & editing. **Prabhakar Mishra** supervised the study, conceptualized the methodology, and contributed to writing – review & editing.

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Declarations

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